SIMATS ENGINEERING
ASSIGNMENT TOOL-4

COURSE NAME: Fundamentals of Data Science **COURSE CODE:** DSA0402

COURSE OUTCOME COVERED

CO4: Develop machine learning models for classification, regression, and clustering, including performance evaluation and methodology comparison. (BL3)

Assessment Tool Weightage: 30

QUESTIONS: 01 **Total Marks:100**

Annexure A

Scenario-Based Assessment

Code of Conduct:

*I **Jannathul Firdaus. N (192573025)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Jannathul Firdaus. N

RUBRICS FOR CALCULATION

SCENARIO BASED ASSESSMENT							
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	Level	Marks
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario		
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues		
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts		
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided		
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand		

Annexure A

NOTE:

For every question, answer the following:

a) Identify the numerical and binary features used for KNN classification.
b) Calculate the **Euclidean distance** between the new sample and all training samples
c) Calculate the **Manhattan distance** between the new sample and all training samples.
d) Calculate the **Minkowski distance** with **p = 3** between the new sample and all training samples.
e) Calculate the **Hamming distance** between the new sample and all training samples using binary features.
f) Identify the **3 nearest neighbours** separately for each distance measure.
g) Predict the class label using majority voting for each distance measure.
h) Compare the results and explain which distance measure is most suitable for the given scenario.

Total = 100 Marks

Question 1: Student Pass or Fail Prediction

A college wants to predict whether a student will Pass or Fail based on academic and activity-related details.

Student	Study Hours	Attendance %	Assignment Submitted	Lab Participation	Revision Done	Result
S1	2	60	0	0	0	Fail
S2	3	65	1	0	0	Fail
S3	6	80	1	1	1	Pass
S4	7	85	1	1	1	Pass
S5	4	70	1	0	1	Fail
S6	8	90	1	1	1	Pass

A new student has the following details:

Study Hours	Attendance %	Assignment Submitted	Lab Participation	Revision Done
5	75	1	1	0

Using KNN with K = 3, classify the new student as Pass or Fail using Euclidean, Manhattan, Minkowski, and Hamming distance.

Question 2: Online Customer Purchase Prediction

An e-commerce company wants to predict whether a customer will Buy or Not Buy a product.

Customer	Browsing Time	Items Viewed	Coupon Used	Added to Cart	Returning Customer	Class
C1	10	3	0	0	0	Not Buy

C2	15	4	1	0	0	Not Buy
C3	25	7	1	1	1	Buy
C4	30	8	1	1	1	Buy
C5	18	5	0	1	0	Not Buy
C6	28	9	1	1	1	Buy

A new customer has:

Browsing Time	Items Viewed	Coupon Used	Added to Cart	Returning Customer
22	6	1	1	0

Using KNN with K = 3, classify the new customer as Buy or Not Buy using all four distance measures.

Question 3: Medical Risk Classification

A hospital wants to classify whether a patient has High Risk or Low Risk based on medical measurements and symptoms.

Patient	Age	Blood Sugar Level	Fever	Cough	Fatigue	Class
P1	25	90	0	0	0	Low Risk
P2	30	95	0	1	0	Low Risk
P3	50	160	1	1	1	High Risk
P4	55	170	1	1	1	High Risk
P5	40	120	0	1	1	Low Risk
P6	60	180	1	1	1	High Risk

A new patient has:

Age	Blood Sugar Level	Fever	Cough	Fatigue
45	150	1	1	0

Using KNN with K = 3, classify the patient as High Risk or Low Risk using Euclidean, Manhattan, Minkowski, and Hamming distance.

Question 4: Fruit Classification

A fruit shop wants to classify a fruit as Apple or Orange based on physical and binary characteristics.

Fruit	Weight	Diameter	Red Color	Citrus Smell	Smooth Skin	Class
F1	150	7	1	0	1	Apple
F2	160	8	1	0	1	Apple
F3	180	9	0	1	0	Orange
F4	200	10	0	1	0	Orange

F5	155	7.5	1	0	1	Apple
F6	190	9.5	0	1	0	Orange

A new fruit has:

Weight	Diameter	Red Color	Citrus Smell	Smooth Skin
170	8.5	1	0	1

Using KNN with K = 3, classify the fruit as Apple or Orange using all four distance measures.

Question 5: House Price Category Prediction

A real estate company wants to classify houses as Low Price or High Price.

House	Area in sq.ft	Bedrooms	Parking Available	Near Main Road	Garden Available	Category
H1	800	2	0	0	0	Low Price
H2	1000	2	1	0	0	Low Price
H3	1500	3	1	1	1	High Price
H4	1800	4	1	1	1	High Price
H5	1200	3	0	1	0	Low Price
H6	2000	4	1	1	1	High Price

A new house has:

Area	Bedrooms	Parking Available	Near Main Road	Garden Available
1400	3	1	1	0

Using KNN with K = 3, classify the house as Low Price or High Price using Euclidean, Manhattan, Minkowski, and Hamming distance.

Question 6: Loan Approval Prediction

A bank wants to predict whether a loan application should be Approved or Rejected.

Applicant	Monthly Income	Credit Score	Govt Job	Existing Loan	Past Default	Class
A1	25000	580	0	1	1	Rejected
A2	30000	600	0	1	0	Rejected
Applicant	Monthly Income	Credit Score	Govt Job	Existing Loan	Past Default	Class
A3	60000	750	1	0	0	Approved
A4	70000	800	1	0	0	Approved
A5	40000	650	0	1	0	Rejected

A6	80000	820	1	0	0	Approved

A new applicant has:

Monthly Income	Credit Score	Govt Job	Existing Loan	Past Default
55000	700	1	0	0

Using KNN with K = 3, classify the loan application as Approved or Rejected using all four distance measures.

Question 7: Movie Preference Classification

A movie platform wants to classify a user as an Action Lover or Comedy Lover.

User	Action Rating	Comedy Rating	Watched Superhero Movie	Watched Stand-up Show	Watched Sitcom	Class
U1	5	1	1	0	0	Action Lover
U2	4	2	1	0	0	Action Lover
U3	1	5	0	1	1	Comedy Lover
U4	2	4	0	1	1	Comedy Lover
U5	5	2	1	0	0	Action Lover
U6	1	4	0	1	1	Comedy Lover

A new user has:

Action Rating	Comedy Rating	Watched Superhero Movie	Watched Stand-up Show	Watched Sitcom
3	3	1	1	0

Using KNN with K = 3, classify the new user as Action Lover or Comedy Lover using Euclidean, Manhattan, Minkowski, and Hamming distance.

Question 8: Crop Disease Detection

An agriculture research center wants to classify a plant as Healthy or Diseased.

Plant	Leaf Spot Count	Leaf Color Index	Yellow Leaves	White Fungus	Wilting	Class
L1	2	80	0	0	0	Healthy
L2	3	75	0	0	0	Healthy
L3	12	40	1	1	1	Diseased
Plant	Leaf Spot Count	Leaf Color Index	Yellow Leaves	White Fungus	Wilting	Class
L4	15	35	1	1	1	Diseased

L5	5	70	0	0	1	Healthy
L6	14	38	1	1	1	Diseased

A new plant has:

Leaf Spot Count	Leaf Color Index	Yellow Leaves	White Fungus	Wilting
10	50	1	1	0

Using KNN with K = 3, classify the plant as Healthy or Diseased using all four distance measures.

Question 9: Network Intrusion Detection

A cybersecurity system wants to classify network activity as Normal or Attack.

Record	Packets per Second	Failed Login Attempts	Unknown Port Used	Admin Access Tried	Blacklisted IP	Class
N1	20	0	0	0	0	Normal
N2	30	1	0	0	0	Normal
N3	100	5	1	1	1	Attack
N4	120	6	1	1	1	Attack
N5	40	1	0	0	0	Normal
N6	110	7	1	1	1	Attack

A new network record has:

Packets per Second	Failed Login Attempts	Unknown Port Used	Admin Access Tried	Blacklisted IP
90	4	1	1	0

Using KNN with K = 3, classify the network activity as Normal or Attack using Euclidean, Manhattan, Minkowski, and Hamming distance.

Question 10: Employee Attrition Prediction

A company wants to predict whether an employee is likely to Stay or Leave.

Employee	Experience in Years	Satisfaction Score	Overtime	Promotion Received	Workplace Conflict	Class
E1	2	8	0	1	0	Stay
E2	3	7	0	1	0	Stay
E3	6	3	1	0	1	Leave
Employee	Experience in Years	Satisfaction Score	Overtime	Promotion Received	Workplace Conflict	Class
E4	7	2	1	0	1	Leave

E5	4	6	0	0	0	Stay
E6	8	2	1	0	1	Leave

A new employee has:

Experience in Years	Satisfaction Score	Overtime	Promotion Received	Workplace Conflict
5	4	1	0	1

Using KNN with K = 3, classify the employee as Stay or Leave using all four distance measures.

Question 1: Student Pass or Fail Prediction

The task is to classify a new student as Fail or Pass using KNN with K = 3. The training data and the new sample are shown below.

ID	Study Hours	Attendance %	Assignment Submitted	Lab Participation	Revision Done	Class
S1	2	60	0	0	0	Fail
S2	3	65	1	0	0	Fail
S3	6	80	1	1	1	Pass
S4	7	85	1	1	1	Pass
S5	4	70	1	0	1	Fail
S6	8	90	1	1	1	Pass
New	**5**	**75**	**1**	**1**	**0**	**?**

(a) Numerical and binary features

- **Numerical features:** Study Hours, Attendance %. These are measured on a numeric scale, so the size of the difference is meaningful.
- **Binary features:** Assignment Submitted, Lab Participation, Revision Done. These are coded 0 = No and 1 = Yes, so two samples either match or differ.
- **Class label (target):** Fail / Pass. It is not used in any distance calculation, only in the vote.

(b) Euclidean distance

Sample	Class	Calculation	Distance
S1	Fail	$(2-5)^2 + (60-75)^2 + (0-1)^2 + (0-1)^2 + (0-0)^2$ $= 9 + 225 + 1 + 1 + 0 = 236$	**$\sqrt{236} = 15.3623$**
S2	Fail	$(3-5)^2 + (65-75)^2 + (1-1)^2 + (0-1)^2 + (0-0)^2$ $= 4 + 100 + 0 + 1 + 0 = 105$	**$\sqrt{105} = 10.2470$**
S3	Pass	$(6-5)^2 + (80-75)^2 + (1-1)^2 + (1-1)^2 + (1-0)^2$ $= 1 + 25 + 0 + 0 + 1 = 27$	**$\sqrt{27} = 5.1962$**
S4	Pass	$(7-5)^2 + (85-75)^2 + (1-1)^2 + (1-1)^2 + (1-0)^2$ $= 4 + 100 + 0 + 0 + 1 = 105$	**$\sqrt{105} = 10.2470$**
S5	Fail	$(4-5)^2 + (70-75)^2 + (1-1)^2 + (0-1)^2 + (1-0)^2$ $= 1 + 25 + 0 + 1 + 1 = 28$	**$\sqrt{28} = 5.2915$**
S6	Pass	$(8-5)^2 + (90-75)^2 + (1-1)^2 + (1-1)^2 + (1-0)^2$ $= 9 + 225 + 0 + 0 + 1 = 235$	**$\sqrt{235} = 15.3297$**

(c) Manhattan distance

Sample	Class	Calculation	Distance
S1	Fail	$\lvert 2-5\rvert + \lvert 60-75\rvert + \lvert 0-1\rvert + \lvert 0-1\rvert + \lvert 0-0\rvert$ $= 3 + 15 + 1 + 1 + 0$	**20**
S2	Fail	$\lvert 3-5\rvert + \lvert 65-75\rvert + \lvert 1-1\rvert + \lvert 0-1\rvert + \lvert 0-0\rvert$ $= 2 + 10 + 0 + 1 + 0$	**13**
S3	Pass	$\lvert 6-5\rvert + \lvert 80-75\rvert + \lvert 1-1\rvert + \lvert 1-1\rvert + \lvert 1-0\rvert$ $= 1 + 5 + 0 + 0 + 1$	**7**
S4	Pass	$\lvert 7-5\rvert + \lvert 85-75\rvert + \lvert 1-1\rvert + \lvert 1-1\rvert + \lvert 1-0\rvert$ $= 2 + 10 + 0 + 0 + 1$	**13**
S5	Fail	$\lvert 4-5\rvert + \lvert 70-75\rvert + \lvert 1-1\rvert + \lvert 0-1\rvert + \lvert 1-0\rvert$ $= 1 + 5 + 0 + 1 + 1$	**8**
S6	Pass	$\lvert 8-5\rvert + \lvert 90-75\rvert + \lvert 1-1\rvert + \lvert 1-1\rvert + \lvert 1-0\rvert$ $= 3 + 15 + 0 + 0 + 1$	**19**

(d) Minkowski (p = 3) distance

Sample	Class	Calculation	Distance
S1	Fail	$\|2-5\|^3 + \|60-75\|^3 + \|0-1\|^3 + \|0-1\|^3 + \|0-0\|^3$ $= 27 + 3{,}375 + 1 + 1 + 0 = 3{,}404$	**$\sqrt[3]{3{,}404} = 15.0428$**
S2	Fail	$\|3-5\|^3 + \|65-75\|^3 + \|1-1\|^3 + \|0-1\|^3 + \|0-0\|^3$ $= 8 + 1{,}000 + 0 + 1 + 0 = 1{,}009$	**$\sqrt[3]{1{,}009} = 10.0299$**
S3	Pass	$\|6-5\|^3 + \|80-75\|^3 + \|1-1\|^3 + \|1-1\|^3 + \|1-0\|^3$ $= 1 + 125 + 0 + 0 + 1 = 127$	**$\sqrt[3]{127} = 5.0265$**
S4	Pass	$\|7-5\|^3 + \|85-75\|^3 + \|1-1\|^3 + \|1-1\|^3 + \|1-0\|^3$ $= 8 + 1{,}000 + 0 + 0 + 1 = 1{,}009$	**$\sqrt[3]{1{,}009} = 10.0299$**
S5	Fail	$\|4-5\|^3 + \|70-75\|^3 + \|1-1\|^3 + \|0-1\|^3 + \|1-0\|^3$ $= 1 + 125 + 0 + 1 + 1 = 128$	**$\sqrt[3]{128} = 5.0397$**
S6	Pass	$\|8-5\|^3 + \|90-75\|^3 + \|1-1\|^3 + \|1-1\|^3 + \|1-0\|^3$ $= 27 + 3{,}375 + 0 + 0 + 1 = 3{,}403$	**$\sqrt[3]{3{,}403} = 15.0414$**

(e) Hamming (binary features only) distance

Sample	Class	Calculation	Distance
S1	Fail	Binary values: [0, 0, 0] against [1, 1, 0] Mismatches (Assignment ≠, Lab ≠, Revision =) = 1 + 1 + 0	**2**
S2	Fail	Binary values: [1, 0, 0] against [1, 1, 0] Mismatches (Assignment =, Lab ≠, Revision =) = 0 + 1 + 0	**1**
S3	Pass	Binary values: [1, 1, 1] against [1, 1, 0] Mismatches (Assignment =, Lab =, Revision ≠) = 0 + 0 + 1	**1**
S4	Pass	Binary values: [1, 1, 1] against [1, 1, 0] Mismatches (Assignment =, Lab =, Revision ≠) = 0 + 0 + 1	**1**
S5	Fail	Binary values: [1, 0, 1] against [1, 1, 0] Mismatches (Assignment =, Lab ≠, Revision ≠) = 0 + 1 + 1	**2**
S6	Pass	Binary values: [1, 1, 1] against [1, 1, 0] Mismatches (Assignment =, Lab =, Revision ≠) = 0 + 0 + 1	**1**

(f) Three nearest neighbours for each distance measure

The summary table gives every distance with its rank in brackets. The cells of the 3 nearest neighbours are shaded.

Sample	Class	Euclidean	Manhattan	Minkowski (p = 3)	Hamming
S1	Fail	15.3623 (6)	20 (6)	15.0428 (6)	2 (6)
S2	Fail	10.2470 (3)	13 (3)	10.0299 (3)	1 (2)
S3	Pass	5.1962 (1)	7 (1)	5.0265 (1)	1 (1)
S4	Pass	10.2470 (4)	13 (4)	10.0299 (4)	1 (3)
S5	Fail	5.2915 (2)	8 (2)	5.0397 (2)	2 (5)
S6	Pass	15.3297 (5)	19 (5)	15.0414 (5)	1 (4)

- **Euclidean:** S3 (5.1962, Pass), S5 (5.2915, Fail), S2 (10.2470, Fail)
- **Manhattan:** S3 (7, Pass), S5 (8, Fail), S2 (13, Fail)
- **Minkowski (p = 3):** S3 (5.0265, Pass), S5 (5.0397, Fail), S2 (10.0299, Fail)
- **Hamming:** S3 (1, Pass), S2 (1, Fail), S4 (1, Pass)

(g) Prediction by majority voting (K = 3)

Distance measure	3 nearest neighbours (class)	Votes	Predicted class
Euclidean	S3 (Pass), S5 (Fail), S2 (Fail)	Fail: 2; Pass: 1	**Fail**
Manhattan	S3 (Pass), S5 (Fail), S2 (Fail)	Fail: 2; Pass: 1	**Fail**
Minkowski (p = 3)	S3 (Pass), S5 (Fail), S2 (Fail)	Fail: 2; Pass: 1	**Fail**
Hamming	S3 (Pass), S2 (Fail), S4 (Pass)	Fail: 1; Pass: 2	**Pass**

Tie note (Euclidean): S2 and S4 are tied for third place at the same distance, so the tie rule keeps S2. If S4 were taken instead of S2, the prediction would be Pass.

Tie note (Manhattan): S2 and S4 are tied for third place at the same distance, so the tie rule keeps S2. If S4 were taken instead of S2, the prediction would be Pass.

Tie note (Minkowski (p = 3)): S2 and S4 are tied for third place at the same distance, so the tie rule keeps S2. If S4 were taken instead of S2, the prediction would be Pass.

(h) Comparison of results and most suitable distance measure

Measure	**Predicted class**
Euclidean	**Fail**
Manhattan	**Fail**
Minkowski (p = 3)	**Fail**
Hamming	**Pass**
Min-max scaled Euclidean (check)	**Pass (S3, S2, S4)**

Euclidean, Manhattan and Minkowski (p = 3) agree on the first two neighbours: S3 (Pass) is nearest and S5 (Fail) second. S2 (Fail) and S4 (Pass) then tie exactly for third place (Euclidean 10.2470, Manhattan 13, Minkowski 10.0299). The tie rule keeps S2, giving Fail (2–1), but choosing S4 would give Pass (2–1), so these three results are not stable. Hamming gives Pass, because S3, S2, S4 and S6 all differ from the new student in only one binary feature and three of those four are Pass students.

Attendance % (60–90) has the largest numeric range, so it dominates the raw distances. S5 looks close mainly because its attendance (70) is near 75, even though S5 has no lab participation, which the new student does have. After min-max scaling, the three nearest students become S3, S2 and S4 and the prediction is Pass.

Most suitable measure: Euclidean distance, applied after min-max scaling. The data mix continuous and binary features, so scaling lets every feature contribute fairly. Hamming alone discards study hours and attendance and leaves a four-way tie. Minkowski with p = 3 over-emphasises the single largest gap, which is attendance.

Final decision: The new student is classified as **Pass**. S3, the nearest neighbour under every measure, is a Pass student with the same assignment and lab status and a very similar profile (6 hours, 80%). Hamming and scaled Euclidean both support Pass, while the Fail results from the raw measures depend only on a tie-break between S2 and S4. Confidence is moderate.

Question 2: Online Customer Purchase Prediction

The task is to classify a new customer as Not Buy or Buy using KNN with K = 3. The training data and the new sample are shown below.

ID	Browsing Time	Items Viewed	Coupon Used	Added to Cart	Returning Customer	Class
C1	10	3	0	0	0	Not Buy
C2	15	4	1	0	0	Not Buy
C3	25	7	1	1	1	Buy
C4	30	8	1	1	1	Buy
C5	18	5	0	1	0	Not Buy
C6	28	9	1	1	1	Buy
New	**22**	**6**	**1**	**1**	**0**	**?**

(a) Numerical and binary features

- **Numerical features:** Browsing Time, Items Viewed. These are measured on a numeric scale, so the size of the difference is meaningful.
- **Binary features:** Coupon Used, Added to Cart, Returning Customer. These are coded 0 = No and 1 = Yes, so two samples either match or differ.
- **Class label (target):** Not Buy / Buy. It is not used in any distance calculation, only in the vote.

(b) Euclidean distance

Sample	Class	Calculation	Distance
C1	Not Buy	$(10-22)^2 + (3-6)^2 + (0-1)^2 + (0-1)^2 + (0-0)^2$ $= 144 + 9 + 1 + 1 + 0 = 155$	$\sqrt{155} = 12.4499$
C2	Not Buy	$(15-22)^2 + (4-6)^2 + (1-1)^2 + (0-1)^2 + (0-0)^2$ $= 49 + 4 + 0 + 1 + 0 = 54$	$\sqrt{54} = 7.3485$
C3	Buy	$(25-22)^2 + (7-6)^2 + (1-1)^2 + (1-1)^2 + (1-0)^2$ $= 9 + 1 + 0 + 0 + 1 = 11$	$\sqrt{11} = 3.3166$
C4	Buy	$(30-22)^2 + (8-6)^2 + (1-1)^2 + (1-1)^2 + (1-0)^2$ $= 64 + 4 + 0 + 0 + 1 = 69$	$\sqrt{69} = 8.3066$
C5	Not Buy	$(18-22)^2 + (5-6)^2 + (0-1)^2 + (1-1)^2 + (0-0)^2$ $= 16 + 1 + 1 + 0 + 0 = 18$	$\sqrt{18} = 4.2426$
C6	Buy	$(28-22)^2 + (9-6)^2 + (1-1)^2 + (1-1)^2 + (1-0)^2$ $= 36 + 9 + 0 + 0 + 1 = 46$	$\sqrt{46} = 6.7823$

(c) Manhattan distance

Sample	Class	Calculation	Distance
C1	Not Buy	$\lvert10-22\rvert + \lvert3-6\rvert + \lvert0-1\rvert + \lvert0-1\rvert + \lvert0-0\rvert$ $= 12 + 3 + 1 + 1 + 0$	**17**
C2	Not Buy	$\lvert15-22\rvert + \lvert4-6\rvert + \lvert1-1\rvert + \lvert0-1\rvert + \lvert0-0\rvert$ $= 7 + 2 + 0 + 1 + 0$	**10**
C3	Buy	$\lvert25-22\rvert + \lvert7-6\rvert + \lvert1-1\rvert + \lvert1-1\rvert + \lvert1-0\rvert$ $= 3 + 1 + 0 + 0 + 1$	**5**
C4	Buy	$\lvert30-22\rvert + \lvert8-6\rvert + \lvert1-1\rvert + \lvert1-1\rvert + \lvert1-0\rvert$ $= 8 + 2 + 0 + 0 + 1$	**11**
C5	Not Buy	$\lvert18-22\rvert + \lvert5-6\rvert + \lvert0-1\rvert + \lvert1-1\rvert + \lvert0-0\rvert$ $= 4 + 1 + 1 + 0 + 0$	**6**
C6	Buy	$\lvert28-22\rvert + \lvert9-6\rvert + \lvert1-1\rvert + \lvert1-1\rvert + \lvert1-0\rvert$ $= 6 + 3 + 0 + 0 + 1$	**10**

(d) Minkowski (p = 3) distance

Sample	Class	Calculation	Distance
C1	Not Buy	$\|10-22\|^3 + \|3-6\|^3 + \|0-1\|^3 + \|0-1\|^3 + \|0-0\|^3$ $= 1{,}728 + 27 + 1 + 1 + 0 = 1{,}757$	**$\sqrt[3]{1{,}757} = 12.0668$**
C2	Not Buy	$\|15-22\|^3 + \|4-6\|^3 + \|1-1\|^3 + \|0-1\|^3 + \|0-0\|^3$ $= 343 + 8 + 0 + 1 + 0 = 352$	**$\sqrt[3]{352} = 7.0607$**
C3	Buy	$\|25-22\|^3 + \|7-6\|^3 + \|1-1\|^3 + \|1-1\|^3 + \|1-0\|^3$ $= 27 + 1 + 0 + 0 + 1 = 29$	**$\sqrt[3]{29} = 3.0723$**
C4	Buy	$\|30-22\|^3 + \|8-6\|^3 + \|1-1\|^3 + \|1-1\|^3 + \|1-0\|^3$ $= 512 + 8 + 0 + 0 + 1 = 521$	**$\sqrt[3]{521} = 8.0466$**
C5	Not Buy	$\|18-22\|^3 + \|5-6\|^3 + \|0-1\|^3 + \|1-1\|^3 + \|0-0\|^3$ $= 64 + 1 + 1 + 0 + 0 = 66$	**$\sqrt[3]{66} = 4.0412$**
C6	Buy	$\|28-22\|^3 + \|9-6\|^3 + \|1-1\|^3 + \|1-1\|^3 + \|1-0\|^3$ $= 216 + 27 + 0 + 0 + 1 = 244$	**$\sqrt[3]{244} = 6.2488$**

(e) Hamming (binary features only) distance

Sample	Class	Calculation	Distance
C1	Not Buy	Binary values: [0, 0, 0] against [1, 1, 0] Mismatches (Coupon ≠, Added ≠, Returning =) = 1 + 1 + 0	**2**
C2	Not Buy	Binary values: [1, 0, 0] against [1, 1, 0] Mismatches (Coupon =, Added ≠, Returning =) = 0 + 1 + 0	**1**
C3	Buy	Binary values: [1, 1, 1] against [1, 1, 0] Mismatches (Coupon =, Added =, Returning ≠) = 0 + 0 + 1	**1**
C4	Buy	Binary values: [1, 1, 1] against [1, 1, 0] Mismatches (Coupon =, Added =, Returning ≠) = 0 + 0 + 1	**1**
C5	Not Buy	Binary values: [0, 1, 0] against [1, 1, 0] Mismatches (Coupon ≠, Added =, Returning =) = 1 + 0 + 0	**1**
C6	Buy	Binary values: [1, 1, 1] against [1, 1, 0] Mismatches (Coupon =, Added =, Returning ≠) = 0 + 0 + 1	**1**

(f) Three nearest neighbours for each distance measure

The summary table gives every distance with its rank in brackets. The cells of the 3 nearest neighbours are shaded.

Sample	Class	Euclidean	Manhattan	Minkowski (p = 3)	Hamming
C1	Not Buy	12.4499 (6)	17 (6)	12.0668 (6)	2 (6)
C2	Not Buy	7.3485 (4)	10 (4)	7.0607 (4)	1 (4)
C3	Buy	3.3166 (1)	5 (1)	3.0723 (1)	1 (1)
C4	Buy	8.3066 (5)	11 (5)	8.0466 (5)	1 (5)
C5	Not Buy	4.2426 (2)	6 (2)	4.0412 (2)	1 (2)
C6	Buy	6.7823 (3)	10 (3)	6.2488 (3)	1 (3)

- **Euclidean:** C3 (3.3166, Buy), C5 (4.2426, Not Buy), C6 (6.7823, Buy)
- **Manhattan:** C3 (5, Buy), C5 (6, Not Buy), C6 (10, Buy)
- **Minkowski (p = 3):** C3 (3.0723, Buy), C5 (4.0412, Not Buy), C6 (6.2488, Buy)
- **Hamming:** C3 (1, Buy), C5 (1, Not Buy), C6 (1, Buy)

(g) Prediction by majority voting (K = 3)

Distance measure	3 nearest neighbours (class)	Votes	Predicted class
Euclidean	C3 (Buy), C5 (Not Buy), C6 (Buy)	Not Buy: 1; Buy: 2	**Buy**
Manhattan	C3 (Buy), C5 (Not Buy), C6 (Buy)	Not Buy: 1; Buy: 2	**Buy**
Minkowski (p = 3)	C3 (Buy), C5 (Not Buy), C6 (Buy)	Not Buy: 1; Buy: 2	**Buy**
Hamming	C3 (Buy), C5 (Not Buy), C6 (Buy)	Not Buy: 1; Buy: 2	**Buy**

Tie note (Manhattan): C6 and C2 are both at 10; the smaller Euclidean distance (6.7823 against 7.3485) places C6 third. Taking C2 instead would give Not Buy.

(h) Comparison of results and most suitable distance measure

Measure	Predicted class
Euclidean	**Buy**
Manhattan	**Buy**
Minkowski (p = 3)	**Buy**
Hamming	**Buy**
Min-max scaled Euclidean (check)	**Not Buy (C3, C5, C2)**

Euclidean (C3 Buy 3.3166, C5 Not Buy 4.2426, C6 Buy 6.7823) and Minkowski (3.0723, 4.0412, 6.2488) both predict Buy. Manhattan ranks C3 (5) and C5 (6) first, then C2 and C6 tie at 10; the Euclidean tie-break prefers C6 (6.7823 against 7.3485), giving Buy, whereas choosing C2 would give Not Buy. Hamming leaves five customers (C3, C5, C6, C2, C4) at distance 1, so the Euclidean tie-break selects C3, C5 and C6, giving Buy.

After min-max scaling, the three nearest customers become C3, C5 and C2, which predicts Not Buy. So the result is borderline: C6 and C2 compete for the third neighbour place.

Most suitable measure: Euclidean distance. Browsing time (10–30) and items viewed (3–9) are numeric measures of the same behaviour and are on a comparable order of magnitude, and Euclidean gives a strict ranking with no ties. Manhattan and Hamming both produce ties, and Hamming ignores the two numeric behaviour measures entirely.

Final decision: The new customer is classified as **Buy**. The new customer used a coupon and added the item to the cart, as all Buy customers did, and the nearest customer (C3) is a Buyer. Four of four measures give Buy under the stated tie rule, but the confidence is moderate because the scaled check disagrees.

Question 3: Medical Risk Classification

The task is to classify a new patient as Low Risk or High Risk using KNN with K = 3. The training data and the new sample are shown below.

ID	Age	Blood Sugar Level	Fever	Cough	Fatigue	Class
P1	25	90	0	0	0	Low Risk
P2	30	95	0	1	0	Low Risk
P3	50	160	1	1	1	High Risk
P4	55	170	1	1	1	High Risk
P5	40	120	0	1	1	Low Risk
P6	60	180	1	1	1	High Risk
New	**45**	**150**	**1**	**1**	**0**	**?**

(a) Numerical and binary features

- **Numerical features:** Age, Blood Sugar Level. These are measured on a numeric scale, so the size of the difference is meaningful.
- **Binary features:** Fever, Cough, Fatigue. These are coded 0 = No and 1 = Yes, so two samples either match or differ.
- **Class label (target):** Low Risk / High Risk. It is not used in any distance calculation, only in the vote.

(b) Euclidean distance

Sample	Class	Calculation	Distance
P1	Low Risk	$(25-45)^2 + (90-150)^2 + (0-1)^2 + (0-1)^2 + (0-0)^2$ $= 400 + 3{,}600 + 1 + 1 + 0 = 4{,}002$	$\sqrt{4{,}002} = 63.2614$
P2	Low Risk	$(30-45)^2 + (95-150)^2 + (0-1)^2 + (1-1)^2 + (0-0)^2$ $= 225 + 3{,}025 + 1 + 0 + 0 = 3{,}251$	$\sqrt{3{,}251} = 57.0175$
P3	High Risk	$(50-45)^2 + (160-150)^2 + (1-1)^2 + (1-1)^2 + (1-0)^2$ $= 25 + 100 + 0 + 0 + 1 = 126$	$\sqrt{126} = 11.2250$
P4	High Risk	$(55-45)^2 + (170-150)^2 + (1-1)^2 + (1-1)^2 + (1-0)^2$ $= 100 + 400 + 0 + 0 + 1 = 501$	$\sqrt{501} = 22.3830$
P5	Low Risk	$(40-45)^2 + (120-150)^2 + (0-1)^2 + (1-1)^2 + (1-0)^2$ $= 25 + 900 + 1 + 0 + 1 = 927$	$\sqrt{927} = 30.4467$
P6	High Risk	$(60-45)^2 + (180-150)^2 + (1-1)^2 + (1-1)^2 + (1-0)^2$ $= 225 + 900 + 0 + 0 + 1 = 1{,}126$	$\sqrt{1{,}126} = 33.5559$

(c) Manhattan distance

Sample	Class	Calculation	Distance
P1	Low Risk	$\lvert 25-45\rvert + \lvert 90-150\rvert + \lvert 0-1\rvert + \lvert 0-1\rvert + \lvert 0-0\rvert$ $= 20 + 60 + 1 + 1 + 0$	**82**
P2	Low Risk	$\lvert 30-45\rvert + \lvert 95-150\rvert + \lvert 0-1\rvert + \lvert 1-1\rvert + \lvert 0-0\rvert$ $= 15 + 55 + 1 + 0 + 0$	**71**
P3	High Risk	$\lvert 50-45\rvert + \lvert 160-150\rvert + \lvert 1-1\rvert + \lvert 1-1\rvert + \lvert 1-0\rvert$ $= 5 + 10 + 0 + 0 + 1$	**16**
P4	High Risk	$\lvert 55-45\rvert + \lvert 170-150\rvert + \lvert 1-1\rvert + \lvert 1-1\rvert + \lvert 1-0\rvert$ $= 10 + 20 + 0 + 0 + 1$	**31**
P5	Low Risk	$\lvert 40-45\rvert + \lvert 120-150\rvert + \lvert 0-1\rvert + \lvert 1-1\rvert + \lvert 1-0\rvert$ $= 5 + 30 + 1 + 0 + 1$	**37**
P6	High Risk	$\lvert 60-45\rvert + \lvert 180-150\rvert + \lvert 1-1\rvert + \lvert 1-1\rvert + \lvert 1-0\rvert$ $= 15 + 30 + 0 + 0 + 1$	**46**

(d) Minkowski (p = 3) distance

Sample	Class	Calculation	Distance
P1	Low Risk	$\|25-45\|^3 + \|90-150\|^3 + \|0-1\|^3 + \|0-1\|^3 + \|0-0\|^3$ = 8,000 + 216,000 + 1 + 1 + 0 = 224,002	**$\sqrt[3]{224,002} = 60.7320$**
P2	Low Risk	$\|30-45\|^3 + \|95-150\|^3 + \|0-1\|^3 + \|1-1\|^3 + \|0-0\|^3$ = 3,375 + 166,375 + 1 + 0 + 0 = 169,751	**$\sqrt[3]{169,751} = 55.3695$**
P3	High Risk	$\|50-45\|^3 + \|160-150\|^3 + \|1-1\|^3 + \|1-1\|^3 + \|1-0\|^3$ = 125 + 1,000 + 0 + 0 + 1 = 1,126	**$\sqrt[3]{1,126} = 10.4035$**
P4	High Risk	$\|55-45\|^3 + \|170-150\|^3 + \|1-1\|^3 + \|1-1\|^3 + \|1-0\|^3$ = 1,000 + 8,000 + 0 + 0 + 1 = 9,001	**$\sqrt[3]{9,001} = 20.8016$**
P5	Low Risk	$\|40-45\|^3 + \|120-150\|^3 + \|0-1\|^3 + \|1-1\|^3 + \|1-0\|^3$ = 125 + 27,000 + 1 + 0 + 1 = 27,127	**$\sqrt[3]{27,127} = 30.0470$**
P6	High Risk	$\|60-45\|^3 + \|180-150\|^3 + \|1-1\|^3 + \|1-1\|^3 + \|1-0\|^3$ = 3,375 + 27,000 + 0 + 0 + 1 = 30,376	**$\sqrt[3]{30,376} = 31.2016$**

(e) Hamming (binary features only) distance

Sample	Class	Calculation	Distance
P1	Low Risk	Binary values: [0, 0, 0] against [1, 1, 0] Mismatches (Fever ≠, Cough ≠, Fatigue =) = 1 + 1 + 0	**2**
P2	Low Risk	Binary values: [0, 1, 0] against [1, 1, 0] Mismatches (Fever ≠, Cough =, Fatigue =) = 1 + 0 + 0	**1**
P3	High Risk	Binary values: [1, 1, 1] against [1, 1, 0] Mismatches (Fever =, Cough =, Fatigue ≠) = 0 + 0 + 1	**1**
P4	High Risk	Binary values: [1, 1, 1] against [1, 1, 0] Mismatches (Fever =, Cough =, Fatigue ≠) = 0 + 0 + 1	**1**
P5	Low Risk	Binary values: [0, 1, 1] against [1, 1, 0] Mismatches (Fever ≠, Cough =, Fatigue ≠) = 1 + 0 + 1	**2**
P6	High Risk	Binary values: [1, 1, 1] against [1, 1, 0] Mismatches (Fever =, Cough =, Fatigue ≠) = 0 + 0 + 1	**1**

(f) Three nearest neighbours for each distance measure

The summary table gives every distance with its rank in brackets. The cells of the 3 nearest neighbours are shaded.

Sample	Class	Euclidean	Manhattan	Minkowski (p = 3)	Hamming
P1	Low Risk	63.2614 (6)	82 (6)	60.7320 (6)	2 (6)
P2	Low Risk	57.0175 (5)	71 (5)	55.3695 (5)	1 (4)
P3	High Risk	11.2250 (1)	16 (1)	10.4035 (1)	1 (1)
P4	High Risk	22.3830 (2)	31 (2)	20.8016 (2)	1 (2)
P5	Low Risk	30.4467 (3)	37 (3)	30.0470 (3)	2 (5)
P6	High Risk	33.5559 (4)	46 (4)	31.2016 (4)	1 (3)

- **Euclidean:** P3 (11.2250, High Risk), P4 (22.3830, High Risk), P5 (30.4467, Low Risk)
- **Manhattan:** P3 (16, High Risk), P4 (31, High Risk), P5 (37, Low Risk)
- **Minkowski (p = 3):** P3 (10.4035, High Risk), P4 (20.8016, High Risk), P5 (30.0470, Low Risk)
- **Hamming:** P3 (1, High Risk), P4 (1, High Risk), P6 (1, High Risk)

(g) Prediction by majority voting (K = 3)

Distance measure	3 nearest neighbours (class)	Votes	Predicted class
Euclidean	P3 (High Risk), P4 (High Risk), P5 (Low Risk)	Low Risk: 1; High Risk: 2	**High Risk**
Manhattan	P3 (High Risk), P4 (High Risk), P5 (Low Risk)	Low Risk: 1; High Risk: 2	**High Risk**
Minkowski (p = 3)	P3 (High Risk), P4 (High Risk), P5 (Low Risk)	Low Risk: 1; High Risk: 2	**High Risk**

Distance measure	3 nearest neighbours (class)	Votes	Predicted class
Hamming	P3 (High Risk), P4 (High Risk), P6 (High Risk)	Low Risk: 0; High Risk: 3	**High Risk**

(h) Comparison of results and most suitable distance measure

Measure	Predicted class
Euclidean	**High Risk**
Manhattan	**High Risk**
Minkowski (p = 3)	**High Risk**
Hamming	**High Risk**
Min-max scaled Euclidean (check)	**High Risk (P3, P4, P6)**

All four measures predict High Risk. Euclidean (P3 11.2250, P4 22.3830, P5 30.4467), Manhattan (16, 31, 37) and Minkowski (10.4035, 20.8016, 30.0470) select the same neighbours, P3, P4 (High Risk) and P5 (Low Risk), for a 2–1 vote. Hamming leaves P3, P4, P6 and P2 at distance 1, and the Euclidean tie-break chooses P3, P4 and P6, a unanimous High Risk. Scaled Euclidean also picks P3, P4 and P6.

Blood sugar (90–180) dominates the raw distances, but it points the same way as the symptoms, so the conclusion does not depend on the measure.

Most suitable measure: Manhattan distance. Clinical measurements such as blood sugar can contain extreme readings, and Manhattan grows linearly with the gap instead of squaring or cubing it, so one extreme value cannot dominate. Each symptom mismatch still adds exactly 1. Blood sugar should still be scaled before use in practice.

Final decision: The new patient is classified as **High Risk**. The patient's blood sugar (150) is close to the High Risk patients (160–180), and they share fever and cough. The result is stable across all four measures and the scaled check, so confidence is high.

Question 4: Fruit Classification

The task is to classify a new fruit as Apple or Orange using KNN with K = 3. The training data and the new sample are shown below.

ID	Weight	Diameter	Red Color	Citrus Smell	Smooth Skin	Class
F1	150	7	1	0	1	Apple
F2	160	8	1	0	1	Apple
F3	180	9	0	1	0	Orange
F4	200	10	0	1	0	Orange
F5	155	7.5	1	0	1	Apple
F6	190	9.5	0	1	0	Orange
New	**170**	**8.5**	**1**	**0**	**1**	**?**

(a) Numerical and binary features

- **Numerical features:** Weight, Diameter. These are measured on a numeric scale, so the size of the difference is meaningful.
- **Binary features:** Red Color, Citrus Smell, Smooth Skin. These are coded 0 = No and 1 = Yes, so two samples either match or differ.
- **Class label (target):** Apple / Orange. It is not used in any distance calculation, only in the vote.

(b) Euclidean distance

Sample	Class	Calculation	Distance
F1	Apple	$(150-170)^2 + (7-8.5)^2 + (1-1)^2 + (0-0)^2 + (1-1)^2$ $= 400 + 2.25 + 0 + 0 + 0 = 402.25$	**$\sqrt{402.25} = 20.0562$**
F2	Apple	$(160-170)^2 + (8-8.5)^2 + (1-1)^2 + (0-0)^2 + (1-1)^2$ $= 100 + 0.25 + 0 + 0 + 0 = 100.25$	**$\sqrt{100.25} = 10.0125$**
F3	Orange	$(180-170)^2 + (9-8.5)^2 + (0-1)^2 + (1-0)^2 + (0-1)^2$ $= 100 + 0.25 + 1 + 1 + 1 = 103.25$	**$\sqrt{103.25} = 10.1612$**
F4	Orange	$(200-170)^2 + (10-8.5)^2 + (0-1)^2 + (1-0)^2 + (0-1)^2$ $= 900 + 2.25 + 1 + 1 + 1 = 905.25$	**$\sqrt{905.25} = 30.0874$**
F5	Apple	$(155-170)^2 + (7.5-8.5)^2 + (1-1)^2 + (0-0)^2 + (1-1)^2$ $= 225 + 1 + 0 + 0 + 0 = 226$	**$\sqrt{226} = 15.0333$**
F6	Orange	$(190-170)^2 + (9.5-8.5)^2 + (0-1)^2 + (1-0)^2 + (0-1)^2$ $= 400 + 1 + 1 + 1 + 1 = 404$	**$\sqrt{404} = 20.0998$**

(c) Manhattan distance

Sample	Class	Calculation	Distance
F1	Apple	$\lvert150-170\rvert + \lvert7-8.5\rvert + \lvert1-1\rvert + \lvert0-0\rvert + \lvert1-1\rvert$ $= 20 + 1.5 + 0 + 0 + 0$	**21.5000**
F2	Apple	$\lvert160-170\rvert + \lvert8-8.5\rvert + \lvert1-1\rvert + \lvert0-0\rvert + \lvert1-1\rvert$ $= 10 + 0.5 + 0 + 0 + 0$	**10.5000**
F3	Orange	$\lvert180-170\rvert + \lvert9-8.5\rvert + \lvert0-1\rvert + \lvert1-0\rvert + \lvert0-1\rvert$ $= 10 + 0.5 + 1 + 1 + 1$	**13.5000**
F4	Orange	$\lvert200-170\rvert + \lvert10-8.5\rvert + \lvert0-1\rvert + \lvert1-0\rvert + \lvert0-1\rvert$ $= 30 + 1.5 + 1 + 1 + 1$	**34.5000**
F5	Apple	$\lvert155-170\rvert + \lvert7.5-8.5\rvert + \lvert1-1\rvert + \lvert0-0\rvert + \lvert1-1\rvert$ $= 15 + 1 + 0 + 0 + 0$	**16**
F6	Orange	$\lvert190-170\rvert + \lvert9.5-8.5\rvert + \lvert0-1\rvert + \lvert1-0\rvert + \lvert0-1\rvert$ $= 20 + 1 + 1 + 1 + 1$	**24**

(d) Minkowski (p = 3) distance

Sample	Class	Calculation	Distance
F1	Apple	$\lvert150-170\rvert^3 + \lvert7-8.5\rvert^3 + \lvert1-1\rvert^3 + \lvert0-0\rvert^3 + \lvert1-1\rvert^3$ $= 8{,}000 + 3.375 + 0 + 0 + 0 = 8003.375$	**$\sqrt[3]{8003.375} = 20.0028$**

Sample	Class	Calculation	Distance
F2	Apple	$\|160-170\|^3 + \|8-8.5\|^3 + \|1-1\|^3 + \|0-0\|^3 + \|1-1\|^3$ $= 1{,}000 + 0.125 + 0 + 0 + 0 = 1000.125$	$\sqrt[3]{1000.125} = 10.0004$
F3	Orange	$\|180-170\|^3 + \|9-8.5\|^3 + \|0-1\|^3 + \|1-0\|^3 + \|0-1\|^3$ $= 1{,}000 + 0.125 + 1 + 1 + 1 = 1003.125$	$\sqrt[3]{1003.125} = 10.0104$
F4	Orange	$\|200-170\|^3 + \|10-8.5\|^3 + \|0-1\|^3 + \|1-0\|^3 + \|0-1\|^3$ $= 27{,}000 + 3.375 + 1 + 1 + 1 = 27006.375$	$\sqrt[3]{27006.375} = 30.0024$
F5	Apple	$\|155-170\|^3 + \|7.5-8.5\|^3 + \|1-1\|^3 + \|0-0\|^3 + \|1-1\|^3$ $= 3{,}375 + 1 + 0 + 0 + 0 = 3{,}376$	$\sqrt[3]{3{,}376} = 15.0015$
F6	Orange	$\|190-170\|^3 + \|9.5-8.5\|^3 + \|0-1\|^3 + \|1-0\|^3 + \|0-1\|^3$ $= 8{,}000 + 1 + 1 + 1 + 1 = 8{,}004$	$\sqrt[3]{8{,}004} = 20.0033$

(e) Hamming (binary features only) distance

Sample	Class	Calculation	Distance
F1	Apple	Binary values: [1, 0, 1] against [1, 0, 1] Mismatches (Red =, Citrus =, Smooth =) = 0 + 0 + 0	**0**
F2	Apple	Binary values: [1, 0, 1] against [1, 0, 1] Mismatches (Red =, Citrus =, Smooth =) = 0 + 0 + 0	**0**
F3	Orange	Binary values: [0, 1, 0] against [1, 0, 1] Mismatches (Red ≠, Citrus ≠, Smooth ≠) = 1 + 1 + 1	**3**
F4	Orange	Binary values: [0, 1, 0] against [1, 0, 1] Mismatches (Red ≠, Citrus ≠, Smooth ≠) = 1 + 1 + 1	**3**
F5	Apple	Binary values: [1, 0, 1] against [1, 0, 1] Mismatches (Red =, Citrus =, Smooth =) = 0 + 0 + 0	**0**
F6	Orange	Binary values: [0, 1, 0] against [1, 0, 1] Mismatches (Red ≠, Citrus ≠, Smooth ≠) = 1 + 1 + 1	**3**

(f) Three nearest neighbours for each distance measure

The summary table gives every distance with its rank in brackets. The cells of the 3 nearest neighbours are shaded.

Sample	Class	Euclidean	Manhattan	Minkowski (p = 3)	Hamming
F1	Apple	20.0562 (4)	21.5000 (4)	20.0028 (4)	0 (3)
F2	Apple	10.0125 (1)	10.5000 (1)	10.0004 (1)	0 (1)
F3	Orange	10.1612 (2)	13.5000 (2)	10.0104 (2)	3 (4)
F4	Orange	30.0874 (6)	34.5000 (6)	30.0024 (6)	3 (6)
F5	Apple	15.0333 (3)	16 (3)	15.0015 (3)	0 (2)
F6	Orange	20.0998 (5)	24 (5)	20.0033 (5)	3 (5)

- **Euclidean:** F2 (10.0125, Apple), F3 (10.1612, Orange), F5 (15.0333, Apple)
- **Manhattan:** F2 (10.5000, Apple), F3 (13.5000, Orange), F5 (16, Apple)
- **Minkowski (p = 3):** F2 (10.0004, Apple), F3 (10.0104, Orange), F5 (15.0015, Apple)
- **Hamming:** F2 (0, Apple), F5 (0, Apple), F1 (0, Apple)

(g) Prediction by majority voting (K = 3)

Distance measure	3 nearest neighbours (class)	Votes	Predicted class
Euclidean	F2 (Apple), F3 (Orange), F5 (Apple)	Apple: 2; Orange: 1	**Apple**
Manhattan	F2 (Apple), F3 (Orange), F5 (Apple)	Apple: 2; Orange: 1	**Apple**
Minkowski (p = 3)	F2 (Apple), F3 (Orange), F5 (Apple)	Apple: 2; Orange: 1	**Apple**
Hamming	F2 (Apple), F5 (Apple), F1 (Apple)	Apple: 3; Orange: 0	**Apple**

(h) Comparison of results and most suitable distance measure

Measure	Predicted class
Euclidean	**Apple**
Manhattan	**Apple**
Minkowski (p = 3)	**Apple**
Hamming	**Apple**
Min-max scaled Euclidean (check)	**Apple (F2, F5, F1)**

All four measures predict Apple, but by different margins. Euclidean (F2 10.0125, F3 10.1612, F5 15.0333), Manhattan (10.5, 13.5, 16) and Minkowski (10.0004, 10.0104, 15.0015) all include F3, an Orange, as the second neighbour, giving Apple 2–1. This happens because weight (about 150–200) dominates the raw distance and the new fruit weighs 170, between the apples (150–160) and the oranges (180–200).

Hamming ignores weight and diameter and looks only at colour, smell and skin: the new fruit matches all three apples exactly (distance 0) and differs from every orange in all three features (distance 3), so it gives a clean 3–0 vote. Scaled Euclidean also selects F2, F5 and F1.

Most suitable measure: Hamming distance. The three binary features (red colour, citrus smell, smooth skin) separate apples from oranges perfectly, with distance 0 against distance 3, while the raw numeric distances are dominated by weight and pull an orange into the neighbour set. Scaled Euclidean is the best alternative.

Final decision: The new fruit is classified as **Apple**. The fruit is red, has no citrus smell and has smooth skin, exactly like F1, F2 and F5. The result is unanimous across all measures and confidence is high.

Question 5: House Price Category Prediction

The task is to classify a new house as Low Price or High Price using KNN with K = 3. The training data and the new sample are shown below.

ID	Area (sq.ft)	Bedrooms	Parking Available	Near Main Road	Garden Available	Class
H1	800	2	0	0	0	Low Price
H2	1,000	2	1	0	0	Low Price
H3	1,500	3	1	1	1	High Price
H4	1,800	4	1	1	1	High Price
H5	1,200	3	0	1	0	Low Price
H6	2,000	4	1	1	1	High Price
New	**1,400**	**3**	**1**	**1**	**0**	**?**

(a) Numerical and binary features

- **Numerical features:** Area (sq.ft), Bedrooms. These are measured on a numeric scale, so the size of the difference is meaningful.
- **Binary features:** Parking Available, Near Main Road, Garden Available. These are coded 0 = No and 1 = Yes, so two samples either match or differ.
- **Class label (target):** Low Price / High Price. It is not used in any distance calculation, only in the vote.

(b) Euclidean distance

Sample	Class	Calculation	Distance
H1	Low Price	$(800-1,400)^2 + (2-3)^2 + (0-1)^2 + (0-1)^2 + (0-0)^2$ $= 360,000 + 1 + 1 + 1 + 0 = 360,003$	**$\sqrt{360,003} = 600.0025$**
H2	Low Price	$(1,000-1,400)^2 + (2-3)^2 + (1-1)^2 + (0-1)^2 + (0-0)^2$ $= 160,000 + 1 + 0 + 1 + 0 = 160,002$	**$\sqrt{160,002} = 400.0025$**
H3	High Price	$(1,500-1,400)^2 + (3-3)^2 + (1-1)^2 + (1-1)^2 + (1-0)^2$ $= 10,000 + 0 + 0 + 0 + 1 = 10,001$	**$\sqrt{10,001} = 100.0050$**
H4	High Price	$(1,800-1,400)^2 + (4-3)^2 + (1-1)^2 + (1-1)^2 + (1-0)^2$ $= 160,000 + 1 + 0 + 0 + 1 = 160,002$	**$\sqrt{160,002} = 400.0025$**
H5	Low Price	$(1,200-1,400)^2 + (3-3)^2 + (0-1)^2 + (1-1)^2 + (0-0)^2$ $= 40,000 + 0 + 1 + 0 + 0 = 40,001$	**$\sqrt{40,001} = 200.0025$**
H6	High Price	$(2,000-1,400)^2 + (4-3)^2 + (1-1)^2 + (1-1)^2 + (1-0)^2$ $= 360,000 + 1 + 0 + 0 + 1 = 360,002$	**$\sqrt{360,002} = 600.0017$**

(c) Manhattan distance

Sample	Class	Calculation	Distance
H1	Low Price	$\lvert 800-1,400\rvert + \lvert 2-3\rvert + \lvert 0-1\rvert + \lvert 0-1\rvert + \lvert 0-0\rvert$ $= 600 + 1 + 1 + 1 + 0$	**603**
H2	Low Price	$\lvert 1,000-1,400\rvert + \lvert 2-3\rvert + \lvert 1-1\rvert + \lvert 0-1\rvert + \lvert 0-0\rvert$ $= 400 + 1 + 0 + 1 + 0$	**402**
H3	High Price	$\lvert 1,500-1,400\rvert + \lvert 3-3\rvert + \lvert 1-1\rvert + \lvert 1-1\rvert + \lvert 1-0\rvert$ $= 100 + 0 + 0 + 0 + 1$	**101**
H4	High Price	$\lvert 1,800-1,400\rvert + \lvert 4-3\rvert + \lvert 1-1\rvert + \lvert 1-1\rvert + \lvert 1-0\rvert$ $= 400 + 1 + 0 + 0 + 1$	**402**
H5	Low Price	$\lvert 1,200-1,400\rvert + \lvert 3-3\rvert + \lvert 0-1\rvert + \lvert 1-1\rvert + \lvert 0-0\rvert$ $= 200 + 0 + 1 + 0 + 0$	**201**
H6	High Price	$\lvert 2,000-1,400\rvert + \lvert 4-3\rvert + \lvert 1-1\rvert + \lvert 1-1\rvert + \lvert 1-0\rvert$ $= 600 + 1 + 0 + 0 + 1$	**602**

(d) Minkowski (p = 3) distance

Sample	Class	Calculation	Distance
H1	Low Price	$\|800-1,400\|^3 + \|2-3\|^3 + \|0-1\|^3 + \|0-1\|^3 + \|0-0\|^3$ $= 216,000,000 + 1 + 1 + 1 + 0 = 216,000,003$	**$\sqrt[3]{216,000,003} = 600$**
H2	Low Price	$\|1,000-1,400\|^3 + \|2-3\|^3 + \|1-1\|^3 + \|0-1\|^3 + \|0-0\|^3$ $= 64,000,000 + 1 + 0 + 1 + 0 = 64,000,002$	**$\sqrt[3]{64,000,002} = 400$**
H3	High Price	$\|1,500-1,400\|^3 + \|3-3\|^3 + \|1-1\|^3 + \|1-1\|^3 + \|1-0\|^3$ $= 1,000,000 + 0 + 0 + 0 + 1 = 1,000,001$	**$\sqrt[3]{1,000,001} = 100$**
H4	High Price	$\|1,800-1,400\|^3 + \|4-3\|^3 + \|1-1\|^3 + \|1-1\|^3 + \|1-0\|^3$ $= 64,000,000 + 1 + 0 + 0 + 1 = 64,000,002$	**$\sqrt[3]{64,000,002} = 400$**
H5	Low Price	$\|1,200-1,400\|^3 + \|3-3\|^3 + \|0-1\|^3 + \|1-1\|^3 + \|0-0\|^3$ $= 8,000,000 + 0 + 1 + 0 + 0 = 8,000,001$	**$\sqrt[3]{8,000,001} = 200$**
H6	High Price	$\|2,000-1,400\|^3 + \|4-3\|^3 + \|1-1\|^3 + \|1-1\|^3 + \|1-0\|^3$ $= 216,000,000 + 1 + 0 + 0 + 1 = 216,000,002$	**$\sqrt[3]{216,000,002} = 600$**

(e) Hamming (binary features only) distance

Sample	Class	Calculation	Distance
H1	Low Price	Binary values: [0, 0, 0] against [1, 1, 0] Mismatches (Parking ≠, Near ≠, Garden =) = 1 + 1 + 0	**2**
H2	Low Price	Binary values: [1, 0, 0] against [1, 1, 0] Mismatches (Parking =, Near ≠, Garden =) = 0 + 1 + 0	**1**
H3	High Price	Binary values: [1, 1, 1] against [1, 1, 0] Mismatches (Parking =, Near =, Garden ≠) = 0 + 0 + 1	**1**
H4	High Price	Binary values: [1, 1, 1] against [1, 1, 0] Mismatches (Parking =, Near =, Garden ≠) = 0 + 0 + 1	**1**
H5	Low Price	Binary values: [0, 1, 0] against [1, 1, 0] Mismatches (Parking ≠, Near =, Garden =) = 1 + 0 + 0	**1**
H6	High Price	Binary values: [1, 1, 1] against [1, 1, 0] Mismatches (Parking =, Near =, Garden ≠) = 0 + 0 + 1	**1**

(f) Three nearest neighbours for each distance measure

The summary table gives every distance with its rank in brackets. The cells of the 3 nearest neighbours are shaded.

Sample	Class	Euclidean	Manhattan	Minkowski (p = 3)	Hamming
H1	Low Price	600.0025 (6)	603 (6)	600 (6)	2 (6)
H2	Low Price	400.0025 (3)	402 (3)	400 (3)	1 (3)
H3	High Price	100.0050 (1)	101 (1)	100 (1)	1 (1)
H4	High Price	400.0025 (4)	402 (4)	400 (4)	1 (4)
H5	Low Price	200.0025 (2)	201 (2)	200 (2)	1 (2)
H6	High Price	600.0017 (5)	602 (5)	600 (5)	1 (5)

- **Euclidean:** H3 (100.0050, High Price), H5 (200.0025, Low Price), H2 (400.0025, Low Price)
- **Manhattan:** H3 (101, High Price), H5 (201, Low Price), H2 (402, Low Price)
- **Minkowski (p = 3):** H3 (100, High Price), H5 (200, Low Price), H2 (400, Low Price)
- **Hamming:** H3 (1, High Price), H5 (1, Low Price), H2 (1, Low Price)

(g) Prediction by majority voting (K = 3)

Distance measure	3 nearest neighbours (class)	Votes	Predicted class
Euclidean	H3 (High Price), H5 (Low Price), H2 (Low Price)	Low Price: 2; High Price: 1	**Low Price**
Manhattan	H3 (High Price), H5 (Low Price), H2 (Low Price)	Low Price: 2; High Price: 1	**Low Price**
Minkowski (p = 3)	H3 (High Price), H5 (Low Price), H2 (Low Price)	Low Price: 2; High Price: 1	**Low Price**

Distance measure	3 nearest neighbours (class)	Votes	Predicted class
Hamming	H3 (High Price), H5 (Low Price), H2 (Low Price)	Low Price: 2; High Price: 1	**Low Price**

Tie note (Euclidean): H2 and H4 are tied for third place at the same distance, so the tie rule keeps H2. If H4 were taken instead of H2, the prediction would be High Price.

Tie note (Manhattan): H2 and H4 are tied for third place at the same distance, so the tie rule keeps H2. If H4 were taken instead of H2, the prediction would be High Price.

Tie note (Minkowski (p = 3)): H2 and H4 are tied for third place at the same distance, so the tie rule keeps H2. If H4 were taken instead of H2, the prediction would be High Price.

Tie note (Hamming): H2 and H4 are tied for third place at the same distance, so the tie rule keeps H2. If H4 were taken instead of H2, the prediction would be High Price.

(h) Comparison of results and most suitable distance measure

Measure	Predicted class
Euclidean	**Low Price**
Manhattan	**Low Price**
Minkowski (p = 3)	**Low Price**
Hamming	**Low Price**
Min-max scaled Euclidean (check)	**Low Price (H3, H5, H2)**

Euclidean (H3 100.0050, H5 200.0025, H2 400.0025), Manhattan (101, 201, 402) and Minkowski (100, 200, 400) all find H3 (High Price) and H5 (Low Price) first, then H2 and H4 tie exactly for third place. The tie rule keeps H2 (Low Price), giving Low Price 2–1; choosing H4 (High Price) would give High Price 2–1. Hamming leaves five houses at distance 1, and the Euclidean tie-break again reaches the H2/H4 tie, so it also gives Low Price under the tie rule. Scaled Euclidean selects H3, H5 and H2, giving Low Price.

Area is in the hundreds, so in the raw distances bedrooms and amenities only act as tie-breakers. The new house (1,400 sq ft) lies between H5 (1,200, Low) and H3 (1,500, High), which is why the case is borderline.

Most suitable measure: Euclidean distance on scaled data. Raw Euclidean, Manhattan and Minkowski effectively reduce to the area difference, and Hamming alone cannot discriminate (five of six houses tie), so scaling is needed for bedrooms and amenities to matter. Euclidean also gives the clearest ranking of the first two neighbours.

Final decision: The new house is classified as **Low Price**. Every measure, including the scaled check, gives Low Price under the stated tie rule, but the third neighbour is a genuine tie (H2 Low against H4 High). A larger K or more training data would be needed to settle the borderline case.

Question 6: Loan Approval Prediction

The task is to classify a new applicant as Rejected or Approved using KNN with K = 3. The training data and the new sample are shown below.

ID	Monthly Income	Credit Score	Govt Job	Existing Loan	Past Default	Class
A1	25,000	580	0	1	1	Rejected
A2	30,000	600	0	1	0	Rejected
A3	60,000	750	1	0	0	Approved
A4	70,000	800	1	0	0	Approved
A5	40,000	650	0	1	0	Rejected
A6	80,000	820	1	0	0	Approved
New	**55,000**	**700**	**1**	**0**	**0**	**?**

(a) Numerical and binary features

- **Numerical features:** Monthly Income, Credit Score. These are measured on a numeric scale, so the size of the difference is meaningful.
- **Binary features:** Govt Job, Existing Loan, Past Default. These are coded 0 = No and 1 = Yes, so two samples either match or differ.
- **Class label (target):** Rejected / Approved. It is not used in any distance calculation, only in the vote.

(b) Euclidean distance

Sample	Class	Calculation	Distance
A1	Rejected	$(25{,}000-55{,}000)^2 + (580-700)^2 + (0-1)^2 + (1-0)^2 + (1-0)^2$ $= 900{,}000{,}000 + 14{,}400 + 1 + 1 + 1 = 900{,}014{,}403$	$\sqrt{900{,}014{,}403} = 30000.2400$
A2	Rejected	$(30{,}000-55{,}000)^2 + (600-700)^2 + (0-1)^2 + (1-0)^2 + (0-0)^2$ $= 625{,}000{,}000 + 10{,}000 + 1 + 1 + 0 = 625{,}010{,}002$	$\sqrt{625{,}010{,}002} = 25000.2000$
A3	Approved	$(60{,}000-55{,}000)^2 + (750-700)^2 + (1-1)^2 + (0-0)^2 + (0-0)^2$ $= 25{,}000{,}000 + 2{,}500 + 0 + 0 + 0 = 25{,}002{,}500$	$\sqrt{25{,}002{,}500} = 5000.2500$
A4	Approved	$(70{,}000-55{,}000)^2 + (800-700)^2 + (1-1)^2 + (0-0)^2 + (0-0)^2$ $= 225{,}000{,}000 + 10{,}000 + 0 + 0 + 0 = 225{,}010{,}000$	$\sqrt{225{,}010{,}000} = 15000.3333$
A5	Rejected	$(40{,}000-55{,}000)^2 + (650-700)^2 + (0-1)^2 + (1-0)^2 + (0-0)^2$ $= 225{,}000{,}000 + 2{,}500 + 1 + 1 + 0 = 225{,}002{,}502$	$\sqrt{225{,}002{,}502} = 15000.0834$
A6	Approved	$(80{,}000-55{,}000)^2 + (820-700)^2 + (1-1)^2 + (0-0)^2 + (0-0)^2$ $= 625{,}000{,}000 + 14{,}400 + 0 + 0 + 0 = 625{,}014{,}400$	$\sqrt{625{,}014{,}400} = 25000.2880$

(c) Manhattan distance

Sample	Class	Calculation	Distance
A1	Rejected	$\lvert 25{,}000-55{,}000\rvert + \lvert 580-700\rvert + \lvert 0-1\rvert + \lvert 1-0\rvert + \lvert 1-0\rvert$ $= 30{,}000 + 120 + 1 + 1 + 1$	**30,123**
A2	Rejected	$\lvert 30{,}000-55{,}000\rvert + \lvert 600-700\rvert + \lvert 0-1\rvert + \lvert 1-0\rvert + \lvert 0-0\rvert$ $= 25{,}000 + 100 + 1 + 1 + 0$	**25,102**
A3	Approved	$\lvert 60{,}000-55{,}000\rvert + \lvert 750-700\rvert + \lvert 1-1\rvert + \lvert 0-0\rvert + \lvert 0-0\rvert$ $= 5{,}000 + 50 + 0 + 0 + 0$	**5,050**
A4	Approved	$\lvert 70{,}000-55{,}000\rvert + \lvert 800-700\rvert + \lvert 1-1\rvert + \lvert 0-0\rvert + \lvert 0-0\rvert$ $= 15{,}000 + 100 + 0 + 0 + 0$	**15,100**
A5	Rejected	$\lvert 40{,}000-55{,}000\rvert + \lvert 650-700\rvert + \lvert 0-1\rvert + \lvert 1-0\rvert + \lvert 0-0\rvert$ $= 15{,}000 + 50 + 1 + 1 + 0$	**15,052**
A6	Approved	$\lvert 80{,}000-55{,}000\rvert + \lvert 820-700\rvert + \lvert 1-1\rvert + \lvert 0-0\rvert + \lvert 0-0\rvert$ $= 25{,}000 + 120 + 0 + 0 + 0$	**25,120**

(d) Minkowski (p = 3) distance

Sample	Class	Calculation	Distance
A1	Rejected	$\lvert 25{,}000-55{,}000\rvert^3 + \lvert 580-700\rvert^3 + \lvert 0-1\rvert^3 + \lvert 1-0\rvert^3 + \lvert 1-0\rvert^3$ $= 27{,}000{,}000{,}000{,}000 + 1{,}728{,}000 + 1 + 1 + 1 = 27{,}000{,}001{,}728{,}003$	$\sqrt[3]{27{,}000{,}001{,}728{,}003} = 30000.0006$

Sample	Class	Calculation	Distance
A2	Rejected	$\|30,000-55,000\|^3 + \|600-700\|^3 + \|0-1\|^3 + \|1-0\|^3 + \|0-0\|^3$ $= 15,625,000,000,000 + 1,000,000 + 1 + 1 + 0 = 15,625,001,000,002$	$\sqrt[3]{15,625,001,000,002} = 25000.0005$
A3	Approved	$\|60,000-55,000\|^3 + \|750-700\|^3 + \|1-1\|^3 + \|0-0\|^3 + \|0-0\|^3$ $= 125,000,000,000 + 125,000 + 0 + 0 + 0 = 125,000,125,000$	$\sqrt[3]{125,000,125,000} = 5000.0017$
A4	Approved	$\|70,000-55,000\|^3 + \|800-700\|^3 + \|1-1\|^3 + \|0-0\|^3 + \|0-0\|^3$ $= 3,375,000,000,000 + 1,000,000 + 0 + 0 + 0 = 3,375,001,000,000$	$\sqrt[3]{3,375,001,000,000} = 15000.0015$
A5	Rejected	$\|40,000-55,000\|^3 + \|650-700\|^3 + \|0-1\|^3 + \|1-0\|^3 + \|0-0\|^3$ $= 3,375,000,000,000 + 125,000 + 1 + 1 + 0 = 3,375,000,125,002$	$\sqrt[3]{3,375,000,125,002} = 15000.0002$
A6	Approved	$\|80,000-55,000\|^3 + \|820-700\|^3 + \|1-1\|^3 + \|0-0\|^3 + \|0-0\|^3$ $= 15,625,000,000,000 + 1,728,000 + 0 + 0 + 0 = 15,625,001,728,000$	$\sqrt[3]{15,625,001,728,000} = 25000.0009$

(e) Hamming (binary features only) distance

Sample	Class	Calculation	Distance
A1	Rejected	Binary values: [0, 1, 1] against [1, 0, 0] Mismatches (Govt ≠, Existing ≠, Past ≠) = 1 + 1 + 1	**3**
A2	Rejected	Binary values: [0, 1, 0] against [1, 0, 0] Mismatches (Govt ≠, Existing ≠, Past =) = 1 + 1 + 0	**2**
A3	Approved	Binary values: [1, 0, 0] against [1, 0, 0] Mismatches (Govt =, Existing =, Past =) = 0 + 0 + 0	**0**
A4	Approved	Binary values: [1, 0, 0] against [1, 0, 0] Mismatches (Govt =, Existing =, Past =) = 0 + 0 + 0	**0**
A5	Rejected	Binary values: [0, 1, 0] against [1, 0, 0] Mismatches (Govt ≠, Existing ≠, Past =) = 1 + 1 + 0	**2**
A6	Approved	Binary values: [1, 0, 0] against [1, 0, 0] Mismatches (Govt =, Existing =, Past =) = 0 + 0 + 0	**0**

(f) Three nearest neighbours for each distance measure

The summary table gives every distance with its rank in brackets. The cells of the 3 nearest neighbours are shaded.

Sample	Class	Euclidean	Manhattan	Minkowski (p = 3)	Hamming
A1	Rejected	30000.2400 (6)	30,123 (6)	30000.0006 (6)	3 (6)
A2	Rejected	25000.2000 (4)	25,102 (4)	25000.0005 (4)	2 (5)
A3	Approved	5000.2500 (1)	5,050 (1)	5000.0017 (1)	0 (1)
A4	Approved	15000.3333 (3)	15,100 (3)	15000.0015 (3)	0 (2)
A5	Rejected	15000.0834 (2)	15,052 (2)	15000.0002 (2)	2 (4)
A6	Approved	25000.2880 (5)	25,120 (5)	25000.0009 (5)	0 (3)

- **Euclidean:** A3 (5000.2500, Approved), A5 (15000.0834, Rejected), A4 (15000.3333, Approved)
- **Manhattan:** A3 (5,050, Approved), A5 (15,052, Rejected), A4 (15,100, Approved)
- **Minkowski (p = 3):** A3 (5000.0017, Approved), A5 (15000.0002, Rejected), A4 (15000.0015, Approved)
- **Hamming:** A3 (0, Approved), A4 (0, Approved), A6 (0, Approved)

(g) Prediction by majority voting (K = 3)

Distance measure	3 nearest neighbours (class)	Votes	Predicted class
Euclidean	A3 (Approved), A5 (Rejected), A4 (Approved)	Rejected: 1; Approved: 2	**Approved**
Manhattan	A3 (Approved), A5 (Rejected), A4 (Approved)	Rejected: 1; Approved: 2	**Approved**
Minkowski (p = 3)	A3 (Approved), A5 (Rejected), A4 (Approved)	Rejected: 1; Approved: 2	**Approved**
Hamming	A3 (Approved), A4 (Approved), A6 (Approved)	Rejected: 0; Approved: 3	**Approved**

(h) Comparison of results and most suitable distance measure

Measure	Predicted class
Euclidean	**Approved**
Manhattan	**Approved**
Minkowski (p = 3)	**Approved**
Hamming	**Approved**
Min-max scaled Euclidean (check)	**Approved (A3, A4, A6)**

All four measures predict Approved. Euclidean (A3 5,000.25, A5 15,000.0834, A4 15,000.3333), Manhattan (5,050, 15,052, 15,100) and Minkowski give A3 (Approved), A5 (Rejected) and A4 (Approved), a 2–1 vote. Hamming gives A3, A4 and A6 at distance 0 (the new applicant has a government job, no existing loan and no past default, exactly like all three approved applicants), a 3–0 vote. Scaled Euclidean also selects A3, A4 and A6.

Monthly income is in tens of thousands, so the raw distances are almost entirely the income gap and the credit score and binary features have virtually no effect. That is why A5 (income 40,000, existing loan, Rejected) appears as the second neighbour even though its risk profile differs from the new applicant's.

Most suitable measure: Hamming distance. The government-job, existing-loan and past-default flags carry the risk profile, and the new applicant matches all Approved applicants on them with a perfect 0 against 2–3 for rejected ones. The other three measures reflect only income.

Final decision: The new applicant is classified as **Approved**. The applicant (income 55,000, credit score 700, government job, no loan, no default) matches the Approved pattern on every binary feature and falls between the lower Rejected and higher Approved incomes. All measures and the scaled check agree, so confidence is high.

Question 7: Movie Preference Classification

The task is to classify a new user as Action Lover or Comedy Lover using KNN with K = 3. The training data and the new sample are shown below.

ID	Action Rating	Comedy Rating	Watched Superhero Movie	Watched Stand-up Show	Watched Sitcom	Class
U1	5	1	1	0	0	Action Lover
U2	4	2	1	0	0	Action Lover
U3	1	5	0	1	1	Comedy Lover
U4	2	4	0	1	1	Comedy Lover
U5	5	2	1	0	0	Action Lover
U6	1	4	0	1	1	Comedy Lover
New	**3**	**3**	**1**	**1**	**0**	**?**

(a) Numerical and binary features

- **Numerical features:** Action Rating, Comedy Rating. These are measured on a numeric scale, so the size of the difference is meaningful.
- **Binary features:** Watched Superhero Movie, Watched Stand-up Show, Watched Sitcom. These are coded 0 = No and 1 = Yes, so two samples either match or differ.
- **Class label (target):** Action Lover / Comedy Lover. It is not used in any distance calculation, only in the vote.

(b) Euclidean distance

Sample	Class	Calculation	Distance
U1	Action Lover	$(5-3)^2 + (1-3)^2 + (1-1)^2 + (0-1)^2 + (0-0)^2$ $= 4 + 4 + 0 + 1 + 0 = 9$	$\sqrt{9} = 3$
U2	Action Lover	$(4-3)^2 + (2-3)^2 + (1-1)^2 + (0-1)^2 + (0-0)^2$ $= 1 + 1 + 0 + 1 + 0 = 3$	$\sqrt{3} = 1.7321$
U3	Comedy Lover	$(1-3)^2 + (5-3)^2 + (0-1)^2 + (1-1)^2 + (1-0)^2$ $= 4 + 4 + 1 + 0 + 1 = 10$	$\sqrt{10} = 3.1623$
U4	Comedy Lover	$(2-3)^2 + (4-3)^2 + (0-1)^2 + (1-1)^2 + (1-0)^2$ $= 1 + 1 + 1 + 0 + 1 = 4$	$\sqrt{4} = 2$
U5	Action Lover	$(5-3)^2 + (2-3)^2 + (1-1)^2 + (0-1)^2 + (0-0)^2$ $= 4 + 1 + 0 + 1 + 0 = 6$	$\sqrt{6} = 2.4495$
U6	Comedy Lover	$(1-3)^2 + (4-3)^2 + (0-1)^2 + (1-1)^2 + (1-0)^2$ $= 4 + 1 + 1 + 0 + 1 = 7$	$\sqrt{7} = 2.6458$

(c) Manhattan distance

Sample	Class	Calculation	Distance
U1	Action Lover	$\lvert5-3\rvert + \lvert1-3\rvert + \lvert1-1\rvert + \lvert0-1\rvert + \lvert0-0\rvert$ $= 2 + 2 + 0 + 1 + 0$	**5**
U2	Action Lover	$\lvert4-3\rvert + \lvert2-3\rvert + \lvert1-1\rvert + \lvert0-1\rvert + \lvert0-0\rvert$ $= 1 + 1 + 0 + 1 + 0$	**3**
U3	Comedy Lover	$\lvert1-3\rvert + \lvert5-3\rvert + \lvert0-1\rvert + \lvert1-1\rvert + \lvert1-0\rvert$ $= 2 + 2 + 1 + 0 + 1$	**6**
U4	Comedy Lover	$\lvert2-3\rvert + \lvert4-3\rvert + \lvert0-1\rvert + \lvert1-1\rvert + \lvert1-0\rvert$ $= 1 + 1 + 1 + 0 + 1$	**4**
U5	Action Lover	$\lvert5-3\rvert + \lvert2-3\rvert + \lvert1-1\rvert + \lvert0-1\rvert + \lvert0-0\rvert$ $= 2 + 1 + 0 + 1 + 0$	**4**
U6	Comedy Lover	$\lvert1-3\rvert + \lvert4-3\rvert + \lvert0-1\rvert + \lvert1-1\rvert + \lvert1-0\rvert$ $= 2 + 1 + 1 + 0 + 1$	**5**

(d) Minkowski (p = 3) distance

Sample	Class	Calculation	Distance
U1	Action Lover	$\|5-3\|^3 + \|1-3\|^3 + \|1-1\|^3 + \|0-1\|^3 + \|0-0\|^3$ = 8 + 8 + 0 + 1 + 0 = 17	**$\sqrt[3]{17}$ = 2.5713**
U2	Action Lover	$\|4-3\|^3 + \|2-3\|^3 + \|1-1\|^3 + \|0-1\|^3 + \|0-0\|^3$ = 1 + 1 + 0 + 1 + 0 = 3	**$\sqrt[3]{3}$ = 1.4422**
U3	Comedy Lover	$\|1-3\|^3 + \|5-3\|^3 + \|0-1\|^3 + \|1-1\|^3 + \|1-0\|^3$ = 8 + 8 + 1 + 0 + 1 = 18	**$\sqrt[3]{18}$ = 2.6207**
U4	Comedy Lover	$\|2-3\|^3 + \|4-3\|^3 + \|0-1\|^3 + \|1-1\|^3 + \|1-0\|^3$ = 1 + 1 + 1 + 0 + 1 = 4	**$\sqrt[3]{4}$ = 1.5874**
U5	Action Lover	$\|5-3\|^3 + \|2-3\|^3 + \|1-1\|^3 + \|0-1\|^3 + \|0-0\|^3$ = 8 + 1 + 0 + 1 + 0 = 10	**$\sqrt[3]{10}$ = 2.1544**
U6	Comedy Lover	$\|1-3\|^3 + \|4-3\|^3 + \|0-1\|^3 + \|1-1\|^3 + \|1-0\|^3$ = 8 + 1 + 1 + 0 + 1 = 11	**$\sqrt[3]{11}$ = 2.2240**

(e) Hamming (binary features only) distance

Sample	Class	Calculation	Distance
U1	Action Lover	Binary values: [1, 0, 0] against [1, 1, 0] Mismatches (Watched =, Watched ≠, Watched =) = 0 + 1 + 0	**1**
U2	Action Lover	Binary values: [1, 0, 0] against [1, 1, 0] Mismatches (Watched =, Watched ≠, Watched =) = 0 + 1 + 0	**1**
U3	Comedy Lover	Binary values: [0, 1, 1] against [1, 1, 0] Mismatches (Watched ≠, Watched =, Watched ≠) = 1 + 0 + 1	**2**
U4	Comedy Lover	Binary values: [0, 1, 1] against [1, 1, 0] Mismatches (Watched ≠, Watched =, Watched ≠) = 1 + 0 + 1	**2**
U5	Action Lover	Binary values: [1, 0, 0] against [1, 1, 0] Mismatches (Watched =, Watched ≠, Watched =) = 0 + 1 + 0	**1**
U6	Comedy Lover	Binary values: [0, 1, 1] against [1, 1, 0] Mismatches (Watched ≠, Watched =, Watched ≠) = 1 + 0 + 1	**2**

(f) Three nearest neighbours for each distance measure

The summary table gives every distance with its rank in brackets. The cells of the 3 nearest neighbours are shaded.

Sample	Class	Euclidean	Manhattan	Minkowski (p = 3)	Hamming
U1	Action Lover	3 (5)	5 (5)	2.5713 (5)	1 (3)
U2	Action Lover	1.7321 (1)	3 (1)	1.4422 (1)	1 (1)
U3	Comedy Lover	3.1623 (6)	6 (6)	2.6207 (6)	2 (6)
U4	Comedy Lover	2 (2)	4 (2)	1.5874 (2)	2 (4)
U5	Action Lover	2.4495 (3)	4 (3)	2.1544 (3)	1 (2)
U6	Comedy Lover	2.6458 (4)	5 (4)	2.2240 (4)	2 (5)

- **Euclidean:** U2 (1.7321, Action Lover), U4 (2, Comedy Lover), U5 (2.4495, Action Lover)
- **Manhattan:** U2 (3, Action Lover), U4 (4, Comedy Lover), U5 (4, Action Lover)
- **Minkowski (p = 3):** U2 (1.4422, Action Lover), U4 (1.5874, Comedy Lover), U5 (2.1544, Action Lover)
- **Hamming:** U2 (1, Action Lover), U5 (1, Action Lover), U1 (1, Action Lover)

(g) Prediction by majority voting (K = 3)

Distance measure	3 nearest neighbours (class)	Votes	Predicted class
Euclidean	U2 (Action Lover), U4 (Comedy Lover), U5 (Action Lover)	Action Lover: 2; Comedy Lover: 1	**Action Lover**
Manhattan	U2 (Action Lover), U4 (Comedy Lover), U5 (Action Lover)	Action Lover: 2; Comedy Lover: 1	**Action Lover**
Minkowski (p = 3)	U2 (Action Lover), U4 (Comedy Lover), U5 (Action Lover)	Action Lover: 2; Comedy Lover: 1	**Action Lover**

Distance measure	3 nearest neighbours (class)	Votes	Predicted class
Hamming	U2 (Action Lover), U5 (Action Lover), U1 (Action Lover)	Action Lover: 3; Comedy Lover: 0	**Action Lover**

(h) Comparison of results and most suitable distance measure

Measure	Predicted class
Euclidean	**Action Lover**
Manhattan	**Action Lover**
Minkowski (p = 3)	**Action Lover**
Hamming	**Action Lover**
Min-max scaled Euclidean (check)	**Action Lover (U2, U5, U1)**

All four measures predict Action Lover. Euclidean (U2 1.7321, U4 2.0, U5 2.4495) and Minkowski (1.4422, 1.5874, 2.1544) choose U2 (Action), U4 (Comedy) and U5 (Action), giving 2–1. Manhattan gives U2 (3), then U4 and U5 both at 4, again 2–1. Hamming selects U2, U5 and U1, each differing only in the stand-up feature, giving 3–0. Scaled Euclidean also selects U2, U5 and U1.

The ratings (1–5) are on a similar scale to the 0/1 features, so no feature dominates the distance. The new user's ratings (3, 3) are neutral, so the viewing history (superhero movie and stand-up both watched) decides the outcome.

Most suitable measure: Euclidean distance. All features are on comparable scales, so the raw distances are not distorted and no rescaling is required. Euclidean uses the rating and viewing information together and gives a clearly ordered ranking, whereas Hamming ignores the ratings.

Final decision: The new user is classified as **Action Lover**. Three of the four nearest users are Action Lovers and the new user watched a superhero movie, but the neutral ratings and the stand-up viewing keep the Comedy side close.

Question 8: Crop Disease Detection

The task is to classify a new plant as Healthy or Diseased using KNN with K = 3. The training data and the new sample are shown below.

ID	Leaf Spot Count	Leaf Color Index	Yellow Leaves	White Fungus	Wilting	Class
L1	2	80	0	0	0	Healthy
L2	3	75	0	0	0	Healthy
L3	12	40	1	1	1	Diseased
L4	15	35	1	1	1	Diseased
L5	5	70	0	0	1	Healthy
L6	14	38	1	1	1	Diseased
New	**10**	**50**	**1**	**1**	**0**	**?**

(a) Numerical and binary features

- **Numerical features:** Leaf Spot Count, Leaf Color Index. These are measured on a numeric scale, so the size of the difference is meaningful.
- **Binary features:** Yellow Leaves, White Fungus, Wilting. These are coded 0 = No and 1 = Yes, so two samples either match or differ.
- **Class label (target):** Healthy / Diseased. It is not used in any distance calculation, only in the vote.

(b) Euclidean distance

Sample	Class	Calculation	Distance
L1	Healthy	$(2-10)^2 + (80-50)^2 + (0-1)^2 + (0-1)^2 + (0-0)^2$ $= 64 + 900 + 1 + 1 + 0 = 966$	$\sqrt{966} = 31.0805$
L2	Healthy	$(3-10)^2 + (75-50)^2 + (0-1)^2 + (0-1)^2 + (0-0)^2$ $= 49 + 625 + 1 + 1 + 0 = 676$	$\sqrt{676} = 26$
L3	Diseased	$(12-10)^2 + (40-50)^2 + (1-1)^2 + (1-1)^2 + (1-0)^2$ $= 4 + 100 + 0 + 0 + 1 = 105$	$\sqrt{105} = 10.2470$
L4	Diseased	$(15-10)^2 + (35-50)^2 + (1-1)^2 + (1-1)^2 + (1-0)^2$ $= 25 + 225 + 0 + 0 + 1 = 251$	$\sqrt{251} = 15.8430$
L5	Healthy	$(5-10)^2 + (70-50)^2 + (0-1)^2 + (0-1)^2 + (1-0)^2$ $= 25 + 400 + 1 + 1 + 1 = 428$	$\sqrt{428} = 20.6882$
L6	Diseased	$(14-10)^2 + (38-50)^2 + (1-1)^2 + (1-1)^2 + (1-0)^2$ $= 16 + 144 + 0 + 0 + 1 = 161$	$\sqrt{161} = 12.6886$

(c) Manhattan distance

Sample	Class	Calculation	Distance
L1	Healthy	$\lvert 2-10\rvert + \lvert 80-50\rvert + \lvert 0-1\rvert + \lvert 0-1\rvert + \lvert 0-0\rvert$ $= 8 + 30 + 1 + 1 + 0$	**40**
L2	Healthy	$\lvert 3-10\rvert + \lvert 75-50\rvert + \lvert 0-1\rvert + \lvert 0-1\rvert + \lvert 0-0\rvert$ $= 7 + 25 + 1 + 1 + 0$	**34**
L3	Diseased	$\lvert 12-10\rvert + \lvert 40-50\rvert + \lvert 1-1\rvert + \lvert 1-1\rvert + \lvert 1-0\rvert$ $= 2 + 10 + 0 + 0 + 1$	**13**
L4	Diseased	$\lvert 15-10\rvert + \lvert 35-50\rvert + \lvert 1-1\rvert + \lvert 1-1\rvert + \lvert 1-0\rvert$ $= 5 + 15 + 0 + 0 + 1$	**21**
L5	Healthy	$\lvert 5-10\rvert + \lvert 70-50\rvert + \lvert 0-1\rvert + \lvert 0-1\rvert + \lvert 1-0\rvert$ $= 5 + 20 + 1 + 1 + 1$	**28**
L6	Diseased	$\lvert 14-10\rvert + \lvert 38-50\rvert + \lvert 1-1\rvert + \lvert 1-1\rvert + \lvert 1-0\rvert$ $= 4 + 12 + 0 + 0 + 1$	**17**

(d) Minkowski (p = 3) distance

Sample	Class	Calculation	Distance
L1	Healthy	$\lvert 2-10\rvert^3 + \lvert 80-50\rvert^3 + \lvert 0-1\rvert^3 + \lvert 0-1\rvert^3 + \lvert 0-0\rvert^3$ $= 512 + 27{,}000 + 1 + 1 + 0 = 27{,}514$	$\sqrt[3]{27{,}514} = 30.1892$

Sample	Class	Calculation	Distance
L2	Healthy	$\|3-10\|^3 + \|75-50\|^3 + \|0-1\|^3 + \|0-1\|^3 + \|0-0\|^3$ $= 343 + 15{,}625 + 1 + 1 + 0 = 15{,}970$	**$\sqrt[3]{15{,}970} = 25.1827$**
L3	Diseased	$\|12-10\|^3 + \|40-50\|^3 + \|1-1\|^3 + \|1-1\|^3 + \|1-0\|^3$ $= 8 + 1{,}000 + 0 + 0 + 1 = 1{,}009$	**$\sqrt[3]{1{,}009} = 10.0299$**
L4	Diseased	$\|15-10\|^3 + \|35-50\|^3 + \|1-1\|^3 + \|1-1\|^3 + \|1-0\|^3$ $= 125 + 3{,}375 + 0 + 0 + 1 = 3{,}501$	**$\sqrt[3]{3{,}501} = 15.1844$**
L5	Healthy	$\|5-10\|^3 + \|70-50\|^3 + \|0-1\|^3 + \|0-1\|^3 + \|1-0\|^3$ $= 125 + 8{,}000 + 1 + 1 + 1 = 8{,}128$	**$\sqrt[3]{8{,}128} = 20.1061$**
L6	Diseased	$\|14-10\|^3 + \|38-50\|^3 + \|1-1\|^3 + \|1-1\|^3 + \|1-0\|^3$ $= 64 + 1{,}728 + 0 + 0 + 1 = 1{,}793$	**$\sqrt[3]{1{,}793} = 12.1486$**

(e) Hamming (binary features only) distance

Sample	Class	Calculation	Distance
L1	Healthy	Binary values: [0, 0, 0] against [1, 1, 0] Mismatches (Yellow ≠, White ≠, Wilting =) = 1 + 1 + 0	**2**
L2	Healthy	Binary values: [0, 0, 0] against [1, 1, 0] Mismatches (Yellow ≠, White ≠, Wilting =) = 1 + 1 + 0	**2**
L3	Diseased	Binary values: [1, 1, 1] against [1, 1, 0] Mismatches (Yellow =, White =, Wilting ≠) = 0 + 0 + 1	**1**
L4	Diseased	Binary values: [1, 1, 1] against [1, 1, 0] Mismatches (Yellow =, White =, Wilting ≠) = 0 + 0 + 1	**1**
L5	Healthy	Binary values: [0, 0, 1] against [1, 1, 0] Mismatches (Yellow ≠, White ≠, Wilting ≠) = 1 + 1 + 1	**3**
L6	Diseased	Binary values: [1, 1, 1] against [1, 1, 0] Mismatches (Yellow =, White =, Wilting ≠) = 0 + 0 + 1	**1**

(f) Three nearest neighbours for each distance measure

The summary table gives every distance with its rank in brackets. The cells of the 3 nearest neighbours are shaded.

Sample	Class	Euclidean	Manhattan	Minkowski (p = 3)	Hamming
L1	Healthy	31.0805 (6)	40 (6)	30.1892 (6)	2 (5)
L2	Healthy	26 (5)	34 (5)	25.1827 (5)	2 (4)
L3	Diseased	10.2470 (1)	13 (1)	10.0299 (1)	1 (1)
L4	Diseased	15.8430 (3)	21 (3)	15.1844 (3)	1 (3)
L5	Healthy	20.6882 (4)	28 (4)	20.1061 (4)	3 (6)
L6	Diseased	12.6886 (2)	17 (2)	12.1486 (2)	1 (2)

- **Euclidean:** L3 (10.2470, Diseased), L6 (12.6886, Diseased), L4 (15.8430, Diseased)
- **Manhattan:** L3 (13, Diseased), L6 (17, Diseased), L4 (21, Diseased)
- **Minkowski (p = 3):** L3 (10.0299, Diseased), L6 (12.1486, Diseased), L4 (15.1844, Diseased)
- **Hamming:** L3 (1, Diseased), L6 (1, Diseased), L4 (1, Diseased)

(g) Prediction by majority voting (K = 3)

Distance measure	3 nearest neighbours (class)	Votes	Predicted class
Euclidean	L3 (Diseased), L6 (Diseased), L4 (Diseased)	Healthy: 0; Diseased: 3	**Diseased**
Manhattan	L3 (Diseased), L6 (Diseased), L4 (Diseased)	Healthy: 0; Diseased: 3	**Diseased**
Minkowski (p = 3)	L3 (Diseased), L6 (Diseased), L4 (Diseased)	Healthy: 0; Diseased: 3	**Diseased**
Hamming	L3 (Diseased), L6 (Diseased), L4 (Diseased)	Healthy: 0; Diseased: 3	**Diseased**

(h) Comparison of results and most suitable distance measure

Measure	Predicted class
Euclidean	**Diseased**
Manhattan	**Diseased**
Minkowski (p = 3)	**Diseased**
Hamming	**Diseased**
Min-max scaled Euclidean (check)	**Diseased (L3, L6, L4)**

All four measures predict Diseased, each with a unanimous 3–0 vote. Euclidean (L3 10.2470, L6 12.6886, L4 15.8430), Manhattan (13, 17, 21) and Minkowski (10.0299, 12.1486, 15.1844) choose L3, L6 and L4, all Diseased. Hamming gives the same three plants at distance 1 against 2, 2 and 3 for the healthy plants. Scaled Euclidean selects the same three plants.

The nearest healthy plant (L5) is much farther away (Euclidean 20.6882) than the third diseased plant (15.8430), so the margin is wide. Leaf Color Index (35–80) dominates the raw distances but points the same way as the symptoms.

Most suitable measure: Euclidean distance. Leaf spot count and leaf colour index are continuous physical measurements, and Euclidean keeps the geometric closeness between plants while the binary symptoms add small, consistent contributions. Hamming also separates the classes perfectly and is a good cross-check.

Final decision: The new plant is classified as **Diseased**. The plant shows yellow leaves and white fungus, with a high spot count (10) and a low colour index (50), and all of its nearest neighbours are diseased. All measures and the scaled check agree, so confidence is high.

Question 9: Network Intrusion Detection

The task is to classify a new network record as Normal or Attack using KNN with K = 3. The training data and the new sample are shown below.

ID	Packets per Second	Failed Login Attempts	Unknown Port Used	Admin Access Tried	Blacklisted IP	Class
N1	20	0	0	0	0	Normal
N2	30	1	0	0	0	Normal
N3	100	5	1	1	1	Attack
N4	120	6	1	1	1	Attack
N5	40	1	0	0	0	Normal
N6	110	7	1	1	1	Attack
New	**90**	**4**	**1**	**1**	**0**	**?**

(a) Numerical and binary features

- **Numerical features:** Packets per Second, Failed Login Attempts. These are measured on a numeric scale, so the size of the difference is meaningful.
- **Binary features:** Unknown Port Used, Admin Access Tried, Blacklisted IP. These are coded 0 = No and 1 = Yes, so two samples either match or differ.
- **Class label (target):** Normal / Attack. It is not used in any distance calculation, only in the vote.

(b) Euclidean distance

Sample	Class	Calculation	Distance
N1	Normal	$(20-90)^2 + (0-4)^2 + (0-1)^2 + (0-1)^2 + (0-0)^2$ $= 4{,}900 + 16 + 1 + 1 + 0 = 4{,}918$	**$\sqrt{4{,}918} = 70.1285$**
N2	Normal	$(30-90)^2 + (1-4)^2 + (0-1)^2 + (0-1)^2 + (0-0)^2$ $= 3{,}600 + 9 + 1 + 1 + 0 = 3{,}611$	**$\sqrt{3{,}611} = 60.0916$**
N3	Attack	$(100-90)^2 + (5-4)^2 + (1-1)^2 + (1-1)^2 + (1-0)^2$ $= 100 + 1 + 0 + 0 + 1 = 102$	**$\sqrt{102} = 10.0995$**
N4	Attack	$(120-90)^2 + (6-4)^2 + (1-1)^2 + (1-1)^2 + (1-0)^2$ $= 900 + 4 + 0 + 0 + 1 = 905$	**$\sqrt{905} = 30.0832$**
N5	Normal	$(40-90)^2 + (1-4)^2 + (0-1)^2 + (0-1)^2 + (0-0)^2$ $= 2{,}500 + 9 + 1 + 1 + 0 = 2{,}511$	**$\sqrt{2{,}511} = 50.1099$**
N6	Attack	$(110-90)^2 + (7-4)^2 + (1-1)^2 + (1-1)^2 + (1-0)^2$ $= 400 + 9 + 0 + 0 + 1 = 410$	**$\sqrt{410} = 20.2485$**

(c) Manhattan distance

Sample	Class	Calculation	Distance
N1	Normal	$\lvert 20-90\rvert + \lvert 0-4\rvert + \lvert 0-1\rvert + \lvert 0-1\rvert + \lvert 0-0\rvert$ $= 70 + 4 + 1 + 1 + 0$	**76**
N2	Normal	$\lvert 30-90\rvert + \lvert 1-4\rvert + \lvert 0-1\rvert + \lvert 0-1\rvert + \lvert 0-0\rvert$ $= 60 + 3 + 1 + 1 + 0$	**65**
N3	Attack	$\lvert 100-90\rvert + \lvert 5-4\rvert + \lvert 1-1\rvert + \lvert 1-1\rvert + \lvert 1-0\rvert$ $= 10 + 1 + 0 + 0 + 1$	**12**
N4	Attack	$\lvert 120-90\rvert + \lvert 6-4\rvert + \lvert 1-1\rvert + \lvert 1-1\rvert + \lvert 1-0\rvert$ $= 30 + 2 + 0 + 0 + 1$	**33**
N5	Normal	$\lvert 40-90\rvert + \lvert 1-4\rvert + \lvert 0-1\rvert + \lvert 0-1\rvert + \lvert 0-0\rvert$ $= 50 + 3 + 1 + 1 + 0$	**55**
N6	Attack	$\lvert 110-90\rvert + \lvert 7-4\rvert + \lvert 1-1\rvert + \lvert 1-1\rvert + \lvert 1-0\rvert$ $= 20 + 3 + 0 + 0 + 1$	**24**

(d) Minkowski (p = 3) distance

Sample	Class	Calculation	Distance
N1	Normal	$\|20-90\|^3 + \|0-4\|^3 + \|0-1\|^3 + \|0-1\|^3 + \|0-0\|^3$ $= 343{,}000 + 64 + 1 + 1 + 0 = 343{,}066$	**$\sqrt[3]{343{,}066} = 70.0045$**
N2	Normal	$\|30-90\|^3 + \|1-4\|^3 + \|0-1\|^3 + \|0-1\|^3 + \|0-0\|^3$ $= 216{,}000 + 27 + 1 + 1 + 0 = 216{,}029$	**$\sqrt[3]{216{,}029} = 60.0027$**
N3	Attack	$\|100-90\|^3 + \|5-4\|^3 + \|1-1\|^3 + \|1-1\|^3 + \|1-0\|^3$ $= 1{,}000 + 1 + 0 + 0 + 1 = 1{,}002$	**$\sqrt[3]{1{,}002} = 10.0067$**
N4	Attack	$\|120-90\|^3 + \|6-4\|^3 + \|1-1\|^3 + \|1-1\|^3 + \|1-0\|^3$ $= 27{,}000 + 8 + 0 + 0 + 1 = 27{,}009$	**$\sqrt[3]{27{,}009} = 30.0033$**
N5	Normal	$\|40-90\|^3 + \|1-4\|^3 + \|0-1\|^3 + \|0-1\|^3 + \|0-0\|^3$ $= 125{,}000 + 27 + 1 + 1 + 0 = 125{,}029$	**$\sqrt[3]{125{,}029} = 50.0039$**
N6	Attack	$\|110-90\|^3 + \|7-4\|^3 + \|1-1\|^3 + \|1-1\|^3 + \|1-0\|^3$ $= 8{,}000 + 27 + 0 + 0 + 1 = 8{,}028$	**$\sqrt[3]{8{,}028} = 20.0233$**

(e) Hamming (binary features only) distance

Sample	Class	Calculation	Distance
N1	Normal	Binary values: [0, 0, 0] against [1, 1, 0] Mismatches (Unknown ≠, Admin ≠, Blacklisted =) = 1 + 1 + 0	**2**
N2	Normal	Binary values: [0, 0, 0] against [1, 1, 0] Mismatches (Unknown ≠, Admin ≠, Blacklisted =) = 1 + 1 + 0	**2**
N3	Attack	Binary values: [1, 1, 1] against [1, 1, 0] Mismatches (Unknown =, Admin =, Blacklisted ≠) = 0 + 0 + 1	**1**
N4	Attack	Binary values: [1, 1, 1] against [1, 1, 0] Mismatches (Unknown =, Admin =, Blacklisted ≠) = 0 + 0 + 1	**1**
N5	Normal	Binary values: [0, 0, 0] against [1, 1, 0] Mismatches (Unknown ≠, Admin ≠, Blacklisted =) = 1 + 1 + 0	**2**
N6	Attack	Binary values: [1, 1, 1] against [1, 1, 0] Mismatches (Unknown =, Admin =, Blacklisted ≠) = 0 + 0 + 1	**1**

(f) Three nearest neighbours for each distance measure

The summary table gives every distance with its rank in brackets. The cells of the 3 nearest neighbours are shaded.

Sample	Class	Euclidean	Manhattan	Minkowski (p = 3)	Hamming
N1	Normal	70.1285 (6)	76 (6)	70.0045 (6)	2 (6)
N2	Normal	60.0916 (5)	65 (5)	60.0027 (5)	2 (5)
N3	Attack	10.0995 (1)	12 (1)	10.0067 (1)	1 (1)
N4	Attack	30.0832 (3)	33 (3)	30.0033 (3)	1 (3)
N5	Normal	50.1099 (4)	55 (4)	50.0039 (4)	2 (4)
N6	Attack	20.2485 (2)	24 (2)	20.0233 (2)	1 (2)

- **Euclidean:** N3 (10.0995, Attack), N6 (20.2485, Attack), N4 (30.0832, Attack)
- **Manhattan:** N3 (12, Attack), N6 (24, Attack), N4 (33, Attack)
- **Minkowski (p = 3):** N3 (10.0067, Attack), N6 (20.0233, Attack), N4 (30.0033, Attack)
- **Hamming:** N3 (1, Attack), N6 (1, Attack), N4 (1, Attack)

(g) Prediction by majority voting (K = 3)

Distance measure	3 nearest neighbours (class)	Votes	Predicted class
Euclidean	N3 (Attack), N6 (Attack), N4 (Attack)	Normal: 0; Attack: 3	**Attack**
Manhattan	N3 (Attack), N6 (Attack), N4 (Attack)	Normal: 0; Attack: 3	**Attack**
Minkowski (p = 3)	N3 (Attack), N6 (Attack), N4 (Attack)	Normal: 0; Attack: 3	**Attack**
Hamming	N3 (Attack), N6 (Attack), N4 (Attack)	Normal: 0; Attack: 3	**Attack**

(h) Comparison of results and most suitable distance measure

Measure	Predicted class
Euclidean	**Attack**
Manhattan	**Attack**
Minkowski (p = 3)	**Attack**
Hamming	**Attack**
Min-max scaled Euclidean (check)	**Attack (N3, N4, N6)**

All four measures predict Attack, each with a unanimous 3–0 vote. Euclidean (N3 10.0995, N6 20.2485, N4 30.0832), Manhattan (12, 24, 33) and Minkowski (10.0067, 20.0233, 30.0033) choose N3, N6 and N4. Hamming gives these three records at distance 1 against 2 for every normal record. Scaled Euclidean selects the same attack records.

Packets per second (20–120) dominates the raw distances, but it points the same way as the binary indicators. The nearest normal record (N5) is far away (Euclidean 50.1099), so the margin is wide.

Most suitable measure: Manhattan distance. Network traffic measurements such as packet rate and failed logins often contain bursts, and Manhattan grows linearly with the gap rather than squaring or cubing it, so one traffic spike does not dominate the comparison. Each security flag mismatch still adds exactly 1.

Final decision: The new network record is classified as **Attack**. The record has 90 packets per second, 4 failed logins, an unknown port and an admin access attempt, all typical of the attack records. In intrusion detection a missed attack is costly, and every measure points to Attack, so confidence is high.

Question 10: Employee Attrition Prediction

The task is to classify a new employee as Stay or Leave using KNN with K = 3. The training data and the new sample are shown below.

ID	Experience (Years)	Satisfaction Score	Overtime	Promotion Received	Workplace Conflict	Class
E1	2	8	0	1	0	Stay
E2	3	7	0	1	0	Stay
E3	6	3	1	0	1	Leave
E4	7	2	1	0	1	Leave
E5	4	6	0	0	0	Stay
E6	8	2	1	0	1	Leave
New	**5**	**4**	**1**	**0**	**1**	**?**

(a) Numerical and binary features

- **Numerical features:** Experience (Years), Satisfaction Score. These are measured on a numeric scale, so the size of the difference is meaningful.
- **Binary features:** Overtime, Promotion Received, Workplace Conflict. These are coded 0 = No and 1 = Yes, so two samples either match or differ.
- **Class label (target):** Stay / Leave. It is not used in any distance calculation, only in the vote.

(b) Euclidean distance

Sample	Class	Calculation	Distance
E1	Stay	$(2-5)^2 + (8-4)^2 + (0-1)^2 + (1-0)^2 + (0-1)^2$ $= 9 + 16 + 1 + 1 + 1 = 28$	$\sqrt{28} = 5.2915$
E2	Stay	$(3-5)^2 + (7-4)^2 + (0-1)^2 + (1-0)^2 + (0-1)^2$ $= 4 + 9 + 1 + 1 + 1 = 16$	$\sqrt{16} = 4$
E3	Leave	$(6-5)^2 + (3-4)^2 + (1-1)^2 + (0-0)^2 + (1-1)^2$ $= 1 + 1 + 0 + 0 + 0 = 2$	$\sqrt{2} = 1.4142$
E4	Leave	$(7-5)^2 + (2-4)^2 + (1-1)^2 + (0-0)^2 + (1-1)^2$ $= 4 + 4 + 0 + 0 + 0 = 8$	$\sqrt{8} = 2.8284$
E5	Stay	$(4-5)^2 + (6-4)^2 + (0-1)^2 + (0-0)^2 + (0-1)^2$ $= 1 + 4 + 1 + 0 + 1 = 7$	$\sqrt{7} = 2.6458$
E6	Leave	$(8-5)^2 + (2-4)^2 + (1-1)^2 + (0-0)^2 + (1-1)^2$ $= 9 + 4 + 0 + 0 + 0 = 13$	$\sqrt{13} = 3.6056$

(c) Manhattan distance

Sample	Class	Calculation	Distance
E1	Stay	$\lvert 2-5\rvert + \lvert 8-4\rvert + \lvert 0-1\rvert + \lvert 1-0\rvert + \lvert 0-1\rvert$ $= 3 + 4 + 1 + 1 + 1$	**10**
E2	Stay	$\lvert 3-5\rvert + \lvert 7-4\rvert + \lvert 0-1\rvert + \lvert 1-0\rvert + \lvert 0-1\rvert$ $= 2 + 3 + 1 + 1 + 1$	**8**
E3	Leave	$\lvert 6-5\rvert + \lvert 3-4\rvert + \lvert 1-1\rvert + \lvert 0-0\rvert + \lvert 1-1\rvert$ $= 1 + 1 + 0 + 0 + 0$	**2**
E4	Leave	$\lvert 7-5\rvert + \lvert 2-4\rvert + \lvert 1-1\rvert + \lvert 0-0\rvert + \lvert 1-1\rvert$ $= 2 + 2 + 0 + 0 + 0$	**4**
E5	Stay	$\lvert 4-5\rvert + \lvert 6-4\rvert + \lvert 0-1\rvert + \lvert 0-0\rvert + \lvert 0-1\rvert$ $= 1 + 2 + 1 + 0 + 1$	**5**
E6	Leave	$\lvert 8-5\rvert + \lvert 2-4\rvert + \lvert 1-1\rvert + \lvert 0-0\rvert + \lvert 1-1\rvert$ $= 3 + 2 + 0 + 0 + 0$	**5**

(d) Minkowski (p = 3) distance

Sample	Class	Calculation	Distance
E1	Stay	$\|2-5\|^3 + \|8-4\|^3 + \|0-1\|^3 + \|1-0\|^3 + \|0-1\|^3$ $= 27 + 64 + 1 + 1 + 1 = 94$	**$\sqrt[3]{94} = 4.5468$**
E2	Stay	$\|3-5\|^3 + \|7-4\|^3 + \|0-1\|^3 + \|1-0\|^3 + \|0-1\|^3$ $= 8 + 27 + 1 + 1 + 1 = 38$	**$\sqrt[3]{38} = 3.3620$**
E3	Leave	$\|6-5\|^3 + \|3-4\|^3 + \|1-1\|^3 + \|0-0\|^3 + \|1-1\|^3$ $= 1 + 1 + 0 + 0 + 0 = 2$	**$\sqrt[3]{2} = 1.2599$**
E4	Leave	$\|7-5\|^3 + \|2-4\|^3 + \|1-1\|^3 + \|0-0\|^3 + \|1-1\|^3$ $= 8 + 8 + 0 + 0 + 0 = 16$	**$\sqrt[3]{16} = 2.5198$**
E5	Stay	$\|4-5\|^3 + \|6-4\|^3 + \|0-1\|^3 + \|0-0\|^3 + \|0-1\|^3$ $= 1 + 8 + 1 + 0 + 1 = 11$	**$\sqrt[3]{11} = 2.2240$**
E6	Leave	$\|8-5\|^3 + \|2-4\|^3 + \|1-1\|^3 + \|0-0\|^3 + \|1-1\|^3$ $= 27 + 8 + 0 + 0 + 0 = 35$	**$\sqrt[3]{35} = 3.2711$**

(e) Hamming (binary features only) distance

Sample	Class	Calculation	Distance
E1	Stay	Binary values: [0, 1, 0] against [1, 0, 1] Mismatches (Overtime ≠, Promotion ≠, Workplace ≠) = 1 + 1 + 1	**3**
E2	Stay	Binary values: [0, 1, 0] against [1, 0, 1] Mismatches (Overtime ≠, Promotion ≠, Workplace ≠) = 1 + 1 + 1	**3**
E3	Leave	Binary values: [1, 0, 1] against [1, 0, 1] Mismatches (Overtime =, Promotion =, Workplace =) = 0 + 0 + 0	**0**
E4	Leave	Binary values: [1, 0, 1] against [1, 0, 1] Mismatches (Overtime =, Promotion =, Workplace =) = 0 + 0 + 0	**0**
E5	Stay	Binary values: [0, 0, 0] against [1, 0, 1] Mismatches (Overtime ≠, Promotion =, Workplace ≠) = 1 + 0 + 1	**2**
E6	Leave	Binary values: [1, 0, 1] against [1, 0, 1] Mismatches (Overtime =, Promotion =, Workplace =) = 0 + 0 + 0	**0**

(f) Three nearest neighbours for each distance measure

The summary table gives every distance with its rank in brackets. The cells of the 3 nearest neighbours are shaded.

Sample	Class	Euclidean	Manhattan	Minkowski (p = 3)	Hamming
E1	Stay	5.2915 (6)	10 (6)	4.5468 (6)	3 (6)
E2	Stay	4 (5)	8 (5)	3.3620 (5)	3 (5)
E3	Leave	1.4142 (1)	2 (1)	1.2599 (1)	0 (1)
E4	Leave	2.8284 (3)	4 (2)	2.5198 (3)	0 (2)
E5	Stay	2.6458 (2)	5 (3)	2.2240 (2)	2 (4)
E6	Leave	3.6056 (4)	5 (4)	3.2711 (4)	0 (3)

- **Euclidean:** E3 (1.4142, Leave), E5 (2.6458, Stay), E4 (2.8284, Leave)
- **Manhattan:** E3 (2, Leave), E4 (4, Leave), E5 (5, Stay)
- **Minkowski (p = 3):** E3 (1.2599, Leave), E5 (2.2240, Stay), E4 (2.5198, Leave)
- **Hamming:** E3 (0, Leave), E4 (0, Leave), E6 (0, Leave)

(g) Prediction by majority voting (K = 3)

Distance measure	3 nearest neighbours (class)	Votes	Predicted class
Euclidean	E3 (Leave), E5 (Stay), E4 (Leave)	Stay: 1; Leave: 2	**Leave**
Manhattan	E3 (Leave), E4 (Leave), E5 (Stay)	Stay: 1; Leave: 2	**Leave**
Minkowski (p = 3)	E3 (Leave), E5 (Stay), E4 (Leave)	Stay: 1; Leave: 2	**Leave**
Hamming	E3 (Leave), E4 (Leave), E6 (Leave)	Stay: 0; Leave: 3	**Leave**

Tie note (Manhattan): E5 and E6 are both at 5; the smaller Euclidean distance (2.6458 against 3.6056) places E5 third. Taking E6 instead would give Leave.

(h) Comparison of results and most suitable distance measure

Measure	Predicted class
Euclidean	**Leave**
Manhattan	**Leave**
Minkowski (p = 3)	**Leave**
Hamming	**Leave**
Min-max scaled Euclidean (check)	**Leave (E3, E4, E6)**

All four measures predict Leave. Euclidean (E3 1.4142, E5 2.6458, E4 2.8284) and Minkowski (1.2599, 2.2240, 2.5198) select E3 (Leave), E5 (Stay) and E4 (Leave), a 2–1 vote. Manhattan gives E3 (2), E4 (4) and then E5 and E6 tie at 5; the Euclidean tie-break picks E5, and either choice still gives Leave. Hamming selects E3, E4 and E6 at distance 0, a unanimous 3–0 vote for Leave. Scaled Euclidean selects the same three employees.

E5 (4 years, satisfaction 6) appears close on the numeric features, but it has no overtime and no conflict, which the new employee does have. Hamming exposes that difference.

Most suitable measure: Euclidean distance. Experience (2–8 years) and satisfaction (2–8 on a 10-point scale) are on similar scales, so the raw distance is not distorted, and Euclidean ranks the neighbours without needing rescaling. Hamming is a strong cross-check for the binary working-condition flags.

Final decision: The new employee is classified as **Leave**. The employee works overtime, has workplace conflict, has had no promotion and a low satisfaction score (4), matching the three Leave employees. All measures and the scaled check agree, so confidence is high.